Topics in Optimal Transport Theory

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Preface

The theory of optimal transport originates in a problem posed by Gaspard Monge in 1781, motivated by practical questions in engineering and military logistics. Monge considered the problem of transporting a distribution of mass to another in such a way that the total cost of displacement is minimized, where cost is measured by the distance mass elements are moved. Despite its intuitive formulation, Monge's original problem is highly nonlinear and lacks a general solution framework.

A major breakthrough came in the 1940s with the work of Leonid Kantorovich, who introduced a relaxed formulation of the problem. Instead of seeking a deterministic map transporting mass, Kantorovich considered transport plans, i.e., joint probability measures with prescribed marginals. This convexification of the problem not only ensured existence of solutions under broad conditions but also revealed deep connections with functional analysis and linear programming. Kantorovich's work laid the foundation for what is now known as the modern theory of optimal transport, and earned him the Nobel Prize in Economics in 1975.

In subsequent decades, the theory developed into a rich mathematical subject, with contributions from Brenier, McCann, and others, who uncovered the geometric structure of optimal maps in Euclidean spaces and their relation to convexity and partial differential equations. In particular, Brenier's theorem established that in quadratic cost settings, optimal transport maps in Euclidean space are gradients of convex functions, revealing a deep link between optimal transport and convex analysis.

Notation

$\mathcal{P}(X)$	Probability measures on X
$\mathcal{P}_p(X)$	Probability measures on X with finite p-moment
$\mathcal{T}(\mu,\nu)$	Transport maps from μ to ν
$Cpl(\mu,\nu)$	Couplings between μ and ν
$\Pi(\mu_1,\dots,\mu_n)$	n-marginal probability measures with marginals $\mu_1,\dots,\mu_n$
$Cpl^M(\mu,\nu)$	Martingale couplings between μ and ν
$Cpl_{bc}(\mu,\nu)$	Bicausal couplings between μ and ν
$C_b(X)$	Continuous, bounded functions on X
$C_p(X)$	Continuous functions on X of p-growth
$C_{b,p}(X)$	Continuous, bounded functions of p-growth
$\mathcal{W}_p(\mu,\nu)$	p-Wasserstein distance between μ and ν

1 Optimal Transport (OT)

1.1 Transport Maps

Definition 1.1.1. A Polish space is a completely metrizable and separable topological space (X, τ). The Borel σ-algebra of X is the σ-algebra $\mathcal{B}(X) := \sigma(\tau)$ generated by the topology of the space.

In the following, we will always consider probability measures μ in some Polish space X, and we will denote the space of all probability measures in X with the symbol $\mathcal{P}(X)$.

Definition 1.1.2. The pushforward by a measurable map $T : X \to Y$ is the induced map $T_\# : \mathcal{P}(X) \to \mathcal{P}(Y)$ such that $T_\#\mu = \mu \circ T^{-1}$, or equivalently $(T_\#\mu)(B) := \mu(T^{-1}(B))$ for all $B \in \mathcal{B}(Y)$.

Definition 1.1.3. A transport map between $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$ is a measurable map $T : X \to Y$ such that $T\mu = \nu$.

We will denote the space of the transport maps by $\mathcal{T}(\mu, \nu) := \{T : X \to Y \mid T\mu = \nu\}$. Intuitively, a transport map between two distributions μ and ν moves points' mass from the first distribution to the second one.

Example 1.1.4. Let $\delta_1, \delta_2 \in \mathcal{P}(\mathbb{R})$ be Dirac measures on $\mathbb{R}$, and $T : \mathbb{R} \to \mathbb{R}$ such that $T(x) = 2x$. Then, we have $T\delta_1(B) = \delta_1(T^{-1}(B)) = \delta_2(B)$ since $T(1) = 2$. Hence, $T \in \mathcal{T}(\delta_1, \delta_2)$ is a transport map between δ_1 and δ_2. Intuitively, the map T transports all the mass of δ_1, which is concentrated at the point 1, to the point 2.
More generally, Dirac measures can only be transported to other Dirac measures by any measurable map $T : X \to Y$. Indeed, consider the measure $\delta_x \in \mathcal{P}(X)$, and a measurable map $T : X \to Y$. Then, for all $B \in \mathcal{B}(Y)$, we have

$$T\delta_x(B) = \delta_x(T^{-1}(B)) = \delta_{T(x)}(B).$$

Note that the other mappings $x \neq y \mapsto T(y)$ are not relevant, as they have zero mass.

Remark 1.1.5. In general, $\mathcal{T}(\mu, \nu) = \varnothing$ given $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$. For instance, there exists no transport maps from a Dirac measure $\delta_x \in \mathcal{P}(X)$ to a non Dirac measure ν.

Example 1.1.6.

(i) Let $\nu \in \mathcal{P}(\mathbb{R})$ be a probability measure on the real line $\mathbb{R}$. The quantile function of ν is the map

$$q_\nu : [0, 1] \to \mathbb{R} \text{ such that } q_\nu(u) := \inf\{x \in \mathbb{R} \mid F_\nu(x) \geq u\}.$$

Then, $q_\nu \in \mathcal{T}(\lambda, \nu)$ is a transport map from the Lebesgue measure λ on $[0, 1]$ to the measure ν. Indeed, for any random variable X, the variable $q_X(U) \sim X$ has the same law of X, where $U \sim U(0, 1)$ is a uniform variable on $[0, 1]$.

(ii) Let $\mu \in \mathcal{P}(\mathbb{R})$ be a continuous probability measure on the real line $\mathbb{R}$. The cumulative function of μ is the map

$$F_\mu : \mathbb{R} \to [0, 1] \text{ such that } F_\mu(x) := \mu((-\infty, x]).$$

Then, $F_\mu \in \mathcal{T}(\mu, \lambda)$ is a transport map from μ to the Lebesgue measure λ. Indeed, for any continuous random variable X, the variable $F_X(X) \sim U$ has the same law of a uniform distribution $U \sim U(0, 1)$.

Lemma 1.1.7. *Let $T_1 : X \to Y$ be a transport map from $\mu \in \mathcal{P}(X)$ to $\nu \in \mathcal{P}(Y)$, and $T_2 : Y \to Z$ be a transport map from $\nu \in \mathcal{P}(Y)$ to $\rho \in \mathcal{P}(Z)$. Then, $T_2 \circ T_1 : X \to Z$ is a transport map*

$$T_2 \circ T_1 \in \mathcal{T}(\mu, \rho)$$

Proof. Let $B \in \mathcal{B}(Z)$, then we have

$$\begin{aligned}((T_2 \circ T_1)_{\#}\mu)(B) &= \mu((T_2 \circ T_1)^{-1}(B)) = \mu(T_1^{-1}(T_2^{-1}(B))) = \\ &= \nu(T_2^{-1}(B)) = \text{ since } T_1 \in \mathcal{T}(\mu, \nu) \\ &= \rho(B) \text{ since } T_2 \in \mathcal{T}(\nu, \rho).\end{aligned}$$

□

Proposition 1.1.8. *Let $\mu, \nu \in \mathcal{P}(\mathbb{R})$ be probability measures with μ continuous. Then, the map $q_\nu \circ F_\mu : \mathbb{R} \to \mathbb{R}$, called Monge map, is a transport map from μ to ν, i.e.*

$$q_\nu \circ F_\mu \in \mathcal{T}(\mu, \nu).$$

Proof. Recall that $F_\mu \in \mathcal{T}(\mu, \lambda)$ and $q_\nu \in \mathcal{T}(\lambda, \nu)$ by (1.1.6), where λ is the Lebesgue measure on $[0, 1]$. Then, the thesis follows from (1.1.7). □

Intuitively, the Monge map $q_\nu \circ F_\mu$ transports mass from μ to ν in a monotone way.

In the setting $X = Y = \mathbb{R}^d$, it is particularly easy to prove if a measurable map $T : \mathbb{R}^d \to \mathbb{R}^d$ is a transport map between two given distributions $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$, as shown by the following proposition.

Proposition 1.1.9. *(Monge - Ampére) $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ absolutely continuous respect to λ Lebesgue measure, $T : \mathbb{R}^d \to \mathbb{R}^d$ injective, C^1 and $\mid JT \mid \neq 0$. Then*

$$T \in \mathcal{T}(\mu, \nu) \iff T \textit{ sol. of the Monge-Ampére eq. } \det(JT) \cdot \left(\frac{d\nu}{d\lambda} \circ T\right) = \frac{d\mu}{d\lambda}. \qquad (1)$$

Proof. Suppose that T is a solution of (1), and we prove that $\mu(T^{-1}(B) = \nu(B)$ for all $B \in \mathcal{B}(\mathbb{R}^d)$. Then, it holds,

$$\begin{aligned}\mu(T^{-1}(B)) &= \int_{T^{-1}(B)} \frac{d\mu}{d\lambda}(\underline{x})d\underline{x} = \text{ since } \mu \ll \lambda \\ &= \int_{T^{-1}(B)} \det(JT(\underline{x}))\frac{d\nu}{d\lambda}(T(\underline{x}))d\underline{x} = \\ &= \int_B \frac{d\nu}{d\nu}(\underline{y})d\underline{y} = \quad \text{by change of variable } \underline{y} = T(\underline{x}) \\ &= \nu(B).\end{aligned}$$

Conversely, suppose that T is a transport from μ to ν, and we show that T solves the Monge - Ampére equation. Thus, we have for all $B \in \mathcal{B}(\mathbb{R}^d)$

$$\nu(B) = \int_B \frac{d\nu}{d\lambda}(\underline{y})d\underline{y} = \int_{T^{-1}(B)} \det(JT(\underline{x}))\frac{d\nu}{d\lambda}(T(\underline{x}))d\underline{x}$$

$$\mu(T^{-1}(B)) = \int_{T^{-1}(B)} \frac{d\mu}{d\lambda}(\underline{x})d\underline{x}.$$

Hence, we have for all $B \in \mathcal{B}(\mathbb{R}^d)$

$$\int_{T^{-1}(B)} \det(JT(\underline{x}))\frac{d\nu}{d\lambda}(T(\underline{x}))d\underline{x} = \int_{T^{-1}(B)} \frac{d\mu}{d\lambda}(\underline{x})d\underline{x},$$

which implies that T solves (1) by fundamental theorem of variational calculus. $\square$

1.2 Couplings

We introduce a generalized notion of transport maps which will be useful to define optimal transport problems.

Definition 1.2.1. A coupling of two probabilities $\mu \in \mathcal{P}(X), \nu \in \mathcal{P}(Y)$ is a probability measure $\pi \in \mathcal{P}(X \times Y)$ with marginals μ, ν, i.e. such that $\pi(A \times Y) = \mu(A)$ and $\pi(X \times B) = \nu(B)$ for all $A \in \mathcal{B}(X), B \in \mathcal{B}(Y)$.

We will denote the space of couplings between μ and ν by $Cpl(\mu, \nu)$.
Equivalently, we can define a coupling between μ and ν as a probability measure $\pi \in \mathcal{P}(X \times Y)$ which induces μ, ν by pushforward under projections $p_X : X \times Y \to X$ and $p_Y : X \times Y \to Y$, i.e. such that $p_X\pi = \mu$ and $p_Y\pi = \nu$. Indeed, for $A \in \mathcal{B}(X), B \in \mathcal{B}(Y)$ we have

$$\begin{aligned}(p_X\pi)(A) = \pi(p_X^{-1}(A)) = \pi(A \times Y) = \mu(A)\\ (p_Y\pi)(B) = \pi(p_Y^{-1}(B)) = \pi(X \times B) = \nu(B).\end{aligned}$$

A further equivalent characterization of the coupling property is given by the following proposition.

Proposition 1.2.2. *Let μ, ν be probability measures on X, Y respectively. Then,*

$$Cpl(\mu, \nu) = \left\{\pi \in \mathcal{P}(X \times Y) \mid \int f d\pi = \int f\mu, \int g d\pi = \int g\nu,\ f \in L^1(\mu),\ g \in L^1(\nu)\right\}$$

Proof. Let $\pi \in \mathcal{P}(X \times Y)$ such that $\int_{X\times Y} f d\pi = \int_X f\mu$ and $\int_{X\times Y} g d\pi = \int_Y g\nu$ for any $f \in L^1(\mu), g \in L^1(\nu)$. Then, for all $A \in \mathcal{B}(X)$ we have

$$\pi(A \times Y) = \int_{X\times Y} 1_{A\times Y}(x, y)d\pi(x, y) = \int_{X\times Y} 1_A(x)d\pi(x, y) = \int_X 1_A\mu = \mu(A).$$

Similarly, it holds $\pi(X \times B) = \nu(B)$ for all $B \in \mathcal{B}(Y)$. Then, $\pi \in Cpl(\mu, \nu)$ is a coupling between μ and ν.
Conversely, suppose that $\pi \in Cpl(\mu, \nu)$ is a coupling between μ and ν, and let $f \in L^1(\mu)$ be an integrable function. Then,

$$\begin{aligned}\int_{X\times Y} f(x)d\pi(x, y) &= \int_{p_X^{-1}(X)} f(p_X(x, y))d\pi(x, y) = \\ &= \int_X f(z)d(p_X\pi)(x) = \text{ by substitution with } z = p_X(x, y)\\ &= \int_X f(z)d\mu(z) \text{ since } \pi \in Cpl(\mu, \nu).\end{aligned}$$

Similarly, one can prove that $\int_{X\times Y} g d\pi = \int_Y g d\nu$ for all $g \in L^1(\nu)$. $\square$

Example 1.2.3. Let $\mu, \nu \sim \mathcal{N}(0,1)$ be univariate standard normal distributions. Then, any coupling of μ and ν is a bivariate normal distribution

$$\pi \sim \mathcal{N}\left((0,0)^T, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$$

for some correlation parameter $\rho \in [-1,1]$.

Remark 1.2.4. Intuitively, a coupling of two measures μ, ν encodes a specific dependency structure between the two distribution. In particular, $Cpl(\mu,\nu) \neq \varnothing$ for any couple of probability measures, because we can always consider the product distribution $\mu \otimes \nu \in Cpl(\mu,\nu)$ which encodes the independence structure. Note that, by definition of product measure, it holds

$$(\mu \otimes \nu)(X \times B) = \mu(X)\nu(B) = \nu(B), \text{ and } \nu \otimes \nu(A \times Y) = \mu(A)\nu(Y) = \nu(A).$$

Example 1.2.5. Let δ_x, δ_y be Dirac measures on X, Y respectively. Then, there is a unique coupling between δ_x and δ_y, given by the independent coupling $\delta_x \otimes \delta_y \in Cpl(\delta_x, \delta_y)$. Indeed, any coupling $\pi \in Cpl(\delta_x, \delta_y)$ satisfies

$$\pi((X \setminus \{x\}) \times Y) = \delta_x(X \setminus \{x\}) = 0, \quad \pi(X \times (Y \setminus \{y\})) = \delta_y(Y \setminus \{y\}) = 0$$

Then, the support of π is contained in $\{(x,y)\}$, which implies $\pi = \delta_{(x,y)} = \delta_x \otimes \delta_y$.

Lemma 1.2.6. *Let $f : X \to Y, g : X \to Z$ be measurable maps, and $\mu \in \mathcal{P}(X)$ probability measure on X. Then,*

$$(f,g)_{\#}\mu \in Cpl(f_{\#}\mu, g_{\#}\mu),$$

where $(f,g) : X \to Y \times Z$ such that $(f,g)(x) = (f(x), g(x))$.

Proof. Let $B \in \mathcal{B}(Y)$, and we prove that the distribution induced by projection of $(f,g)_{\#}\mu$ on Y is the measure $f_{\#}\mu$. We have

$$\begin{aligned} p_{Y\#}((f,g)_{\#}\mu)(B) &= ((f,g)_{\#}\mu)(p_Y^{-1}(B)) = \\ &= \mu((f,g)^{-1}(p_Y^{-1}(B))) = \\ &= \mu(f^{-1}(B)) = (f_{\#}\mu)(B) \text{ since } p_Y \circ (f,g) = f. \end{aligned}$$

Similarly, it can be proven that the second marginal of $(f,g)_{\#}\mu$ is the measure $g_{\#}\mu$. □

Proposition 1.2.7. *Let μ, ν be probability measures on X, Y respectively. Then, the map $T \mapsto (Id_X, T)_{\#}\mu$ gives an injection*

$$\mathcal{T}(\mu,\nu) \hookrightarrow Cpl(\mu,\nu).$$

Proof. Let $T : X \to Y$ be a transport map from μ to ν, and consider the measurable map

$$(Id_X, T) : X \to X \times Y \text{ such that } (Id_X, T)(x) := (x, T(x)).$$

Then, the probability measure

$$\pi := (Id_X, T)_{\#}\mu \in Cpl(Id_{\#}\mu, T_{\#}\mu) = Cpl(\mu,\nu)$$

is a coupling between μ and ν by (1.2.6). □

Remark 1.2.8. The mass of the coupling $\pi = (Id_X, T)_{\#}\mu$ is concentrated on the graph of $T : X \to Y$. Equivalently, the coupling induced by transport maps are the laws of the random vectors of the type $(X, T(X))$. Hence, in the analogy where a transport maps moves entirely the mass of a point $x \in X$ to a point $y = T(x) \in Y$, we can interpret couplings as randomized transport maps. Indeed, a coupling $\pi \in Cpl(\mu, \nu)$ can be interpreted as a random map which assigns the map of the point $x \in X$ to a point $y \in Y$ with probability $\pi(Y \in \cdot \mid X = x)$. For this reason, we will sometimes refer to the couplings as transport plans.

Example 1.2.9.

(i) Let $\mu, \nu \in \mathcal{P}(\mathbb{R})$ be probability measures with μ continuous. Then, the Monge transport $q_\nu \circ F_\mu : \mathbb{R} \to \mathbb{R}$ from μ to ν induces via canonical injection $\mathcal{T}(\mu, \nu) \hookrightarrow Cpl(\mu, \nu)$ a transport plan

$$\pi := (Id_{\mathbb{R}}, q_\nu \circ F_\mu)_{\#}\mu \in Cpl(\mu, \nu),$$

called monotone coupling. Intuitively, the monotone coupling between μ and ν assigns the masses in a monotone way, since both q_ν and F_μ are monotone. Moreover, the support of the coupling is a monotone subset of $\mathbb{R}^2$, since

$$(x_1 - x_2)(T(x_1) - T(x_2)) \geq 0$$

for any couple $(x_1, T(x_1))$, $(x_2, T(x_2))$ in the support of π.

(ii) Let $\mu, \nu \in \mathcal{P}(\mathbb{R})$ be probability measures on the real line, and consider the probability measure $\pi := (q_\mu, q_\mu)_{\#}\lambda$ on $\mathbb{R}^2$ induced by pushforward of the Lebesgue measure under the map

$$(q_\mu, q_\nu) : [0, 1] \to \mathbb{R}^2 \text{ such that } (q_\mu, q_\nu)(u) = (q_\mu(u), q_\nu(u))$$

Then, $\pi := (q_\mu, q_\mu)_{\#}\lambda \in Cpl(\mu, \nu)$ is a coupling between μ and ν by Lemma 1.2.6, called comonotone coupling.

Remark 1.2.10. The previous examples suggest a connection between couplings and copulas. Indeed, copulas provide a parametrization of couplings, while optimal transport selects special couplings according to geometric or variational criteria.

1.3 Topological Properties of $\mathcal{T}(\mu, \nu)$, $Cpl(\mu, \nu)$

We have seen that the space of transport maps $\mathcal{T}(\mu, \nu)$ can be interpreted as a subspace of the space of couplings $Cpl(\mu, \nu)$ through the mapping $T \mapsto (Id_X, T)_{\#}\mu$. In the next section, we will define the Kantorovich and the Monge problem as optimization problems on the spaces $\mathcal{T}(\mu, \nu)$, $Cpl(\mu, \nu)$ respectively. Then, we will now investigate some properties of these spaces.

Definition 1.3.1. A sequence of measures $\{\mu_n\}_n \subseteq \mathcal{P}(X)$ converges weakly to a measure $\mu \in \mathcal{P}(X)$ if

$$\int_X f d\mu_n \underset{n\to\infty}{\longrightarrow} \int_X f d\mu$$

for any continuous bounded function $f : X \to \mathbb{R}$, i.e. $f \in C_b(X)$.

The notion of weak convergence induces a topology on the space of probability measures $\mathcal{P}(x)$, called weak topology, where the closed subspaces $C \subseteq \mathcal{P}(X)$ are exactly the ones preserving the limits.

Definition 1.3.2. A family of probability measures $\mathcal{A} \subseteq \mathcal{P}(X)$ is relatively compact respect to the weak topology if the closure $\bar{\mathcal{A}} \subseteq \mathcal{P}(X)$ is compact. Equivalently, $\mathcal{A}$ is relatively compact if each sequence $\{\mu_n\}_n \subseteq \mathcal{A}$ admits a convergent subsequence in $\mathcal{P}(X)$.

Definition 1.3.3. A family of probability measures $\mathcal{A} \subseteq \mathcal{P}(X)$ is tight if for all $\epsilon > 0$ there exists a compact subspace $K_\varepsilon \subseteq \mathcal{P}(X)$ such that

$$\sup_{\mu \in \mathcal{A}} \mu(X \setminus K_\varepsilon) < \varepsilon.$$

The next result gives a practical characterization of the notion of relative compacteness in $\mathcal{P}(X)$, proving that it is indeed equivalent to the tightness.

Theorem 1.3.4. ***(Prokhorov)*** *Let $\mathcal{A} \subseteq \mathcal{P}(X)$ be a family of probability measures, then*

$$\mathcal{A} \subseteq \mathcal{P}(X) \text{ relatively compact resp. weak topology } \iff \mathcal{A} \subseteq \mathcal{P}(X) \text{ tight.}$$

Lemma 1.3.5. *$\mathcal{A}_1 \subseteq \mathcal{P}(X), \mathcal{A}_2 \subseteq \mathcal{P}(Y)$ tight families of probability measures. Then, the family*

$$\mathcal{A} := \{\pi \in \mathcal{P}(X \times Y) \mid p_X\pi \in \mathcal{A}_1,\ p_Y\pi \in \mathcal{A}_2\}$$

is tight.

Proof. Let $\pi \in \mathcal{A}$ with marginals $\mu := p_X\pi \in \mathcal{A}_1$, $\nu := p_Y\pi \in \mathcal{A}_2$, and consider the compact sets $K_1 \subseteq X,\ K_2 \subseteq Y$ such that $\mu(X \setminus K_1) < \varepsilon$, $\nu(Y \setminus K_2) < \varepsilon$ for a fixed $\varepsilon > 0$. Then, for the compact $K_1 \times K_2 \subseteq X \times Y$, we have

$$\begin{aligned}
\pi((X \times Y) \setminus (K_1 \times K_2)) &\leq \pi(((X \setminus K_1) \times Y) \cup (X \times (Y \setminus K_2))) \leq \\
&\leq \pi((X \setminus K_1) \times Y) + \pi(X \times (Y \setminus K_2)) \leq \\
&\leq \mu(X \setminus K_1) + \nu(Y \setminus K_2) < 2\varepsilon \quad \text{since } \pi \in Cpl(\mu, \nu)
\end{aligned}$$

Then, the thesis follows by taking LHS supremum. □

Lemma 1.3.6. *Let $\mu, \nu \in \mathcal{P}(X)$ be probability measures on X such that $a\mu + b\nu \in \mathcal{P}(X)$ for some $a, b \geq 0$. If $f \in L^1(\mu) \cap L^1(\nu)$, then $f \in L^1(a\mu + b\nu)$ with*

$$\int_X f\ d(a\mu + b\nu) = \int_X f\ d\mu + \int_X f\ d\nu.$$

In particular, the map $\mu \mapsto \int_X f\ d\mu$ is linear with respect to convex combinations of measures.

Proof. Firstly, suppose that $f = \sum_{i=1}^n c_i 1_{A_i}$ simple function for some $c_i \geq 0$, and $A_i \in \mathcal{B}(X)$ for $i = 1, \dots, n$. Then, we get

$$\int f\ d(a\mu + b\nu) = \sum_{i=1}^n c_i(a\mu(A_i) + b\nu(A_i)) = a\int f\ d\mu + b\int f\ d\nu$$

Now, suppose that $f = \lim_{n\to\infty} f_n$ with $\{f_n\}_n$ increasing sequence of simple functions. Then, by monotone convergence we obtain

$$\int f \; d(a\mu + b\nu) = \lim_{n\to\infty} \int f_n \; d(a\mu + b\nu) = a \int f \; d\mu + b \int f \; d\nu$$

Finally, we get the thesis by decomposing $f = f^+ - f^-$ and applying the previous step to f^+, f^-. $\square$

The next result states the properties of the space $Cpl(\mu, \nu)$ of couplings between μ and ν.

Proposition 1.3.7. *Let μ, ν be probability measures on X, Y respectively. Then, the following holds:*

(i) $Cpl(\mu, \nu)$ convex.

(ii) $Cpl(\mu, \nu) \subseteq \mathcal{P}(X \times Y)$ compact.

Proof. *(i)* Let $\pi_1, \pi_2 \in Cpl(\mu, \nu)$ be couplings between μ and ν, and we prove that $\alpha\pi_1 + (1-\alpha)\pi_2$ is a coupling between μ and ν. Let $f \in L^1(\mu)$ be an integrable function, then

$$\begin{aligned}
&\int_{X\times Y} f(x) d(\alpha\pi_1 + (1-\alpha)\pi_2)(x,y) = \\
&\quad = \alpha \int_{X\times Y} f(x) d\pi_1(x,y) \; + \; (1-\alpha) \int_{X\times Y} f(x) d\nu(x,y) = \\
&\quad = \alpha \int_X f(x) d\mu(x) \; + \; (1-\alpha) \int_X f(x) d\pi_2(x,y) = \text{ by (1.3.6) and (1.2.2)} \\
&\quad = \int_X f(x) d\mu(x).
\end{aligned}$$

Analogously, the equality holds for the other marginal ν, proving that the probability measure $\alpha\pi_1 + (1-\alpha)\pi_2 \in Cpl(\mu, \nu)$ is a coupling.

(ii) Consider the compact families $\mathcal{A}_1 = \{\mu\} \subseteq \mathcal{P}(X)$ and $\mathcal{A}_2 = \{\nu\} \subseteq \mathcal{P}(Y)$, which are tight by Prokhorov's theorem. Then, the space of coupling between μ and ν

$$Cpl(\mu, \nu) = \{\pi \in \mathcal{P}(X \times Y) \mid p_X\pi = \mu, \; p_Y\pi = \nu\}$$

is tight by Lemma 1.3.5. In particular, $Cpl(\mu, \nu) \subseteq \mathcal{P}(X \times Y)$ is relatively compact respect to the weak topology, by Theorem 1.3.4.

Now, consider a sequence of probability measures $\{\pi_n\}_n \subseteq Cpl(\mu, \nu)$ converging weakly to a probability measure $\pi \in \mathcal{P}(X \times Y)$. We prove that $Cpl(\mu, \nu)$ is closed in $\mathcal{P}(X \times Y)$, i.e. we prove that the limit measure π is a coupling between μ and ν.

Let $f \in C_b(X)$ a continuous and bounded function, then we have

$$\begin{aligned}
\int_{X\times Y} f(x) d\pi(x,y) &= \int_{X\times Y} \tilde{f}(x,y) d\pi(x,y) = \text{ with } \tilde{f}(x,y) := f(x) \\
&= \lim_{n\to\infty} \int_{X\times Y} \tilde{f}(x,y) d\pi_n(x,y) = \text{ since } \tilde{f} \in C_b(X \times Y) \\
&= \lim_{n\to\infty} \int_{X\times Y} f(x) d\mu(x) = \int_X f(x) d\mu(x) \text{ by Proposition 1.2.2.}
\end{aligned}$$

Similarly, one can prove that $\int_{X\times Y} g(x)d\pi(x,y) = \int_Y g(x)d\nu(x)$ for all continuous and bounded $g \in C_b(Y)$. Then, π is a coupling between μ and ν by Proposition 1.2.2, and $Cpl(\mu,\nu)$ is closed. Hence, $Cpl(\mu,\nu)$ is also compact as a closed and relatively compact subspace of $\mathcal{P}(X\times Y)$. □

Remark 1.3.8. In general, the space $\mathcal{T}(\mu,\nu)$ of transport maps from μ to ν does not satisfy any of the properties above.

(i) $\mathcal{T}(\mu,\nu)$ *not convex.*
Consider the measures $\mu := \frac{1}{2}(\delta_{-1}+\delta_1)$, $\nu := \frac{1}{2}(\delta_{-2}+\delta_2) \in \mathcal{P}(\mathbb{R})$, and the measurable maps $T_1, T_2 : \mathbb{R} \to \mathbb{R}$ such that $T_1(x) = 2x$, $T_2(x) = -2x$. Then, we have

$$T_1\mu = \nu, \quad T_2\mu = \nu$$

since $T_1(-1) = -2$, $T_1(1) = 2$ and $T_2(-1) = 2$, $T_2(1) = -2$. However, the convex combination $T := \frac{1}{2}(T_1+T_2)$ is not a transport from μ to ν, since $T(-1) = T(1) = 0$.

(ii) $\mathcal{T}(\mu,\nu)$ *not closed respect to the weak topology.*
Let $\mu = \nu = \lambda$ be the Lebesgue measure on the interval $[0,1]$, and consider the sequence of measurable maps

$$T_n : [0,1] \to [0,1] \text{ such that } T_n(x) = \begin{cases} x & x \in \bigcup_{k\in\mathbb{N}} [\frac{2k}{2n}, \frac{2k+1}{2n}] \\ 1-x & x \in \bigcup_{k\in\mathbb{N}} [\frac{2k+1}{2n}, \frac{2k+2}{2n}] \end{cases}$$

Then, each $T_n \in \mathcal{T}(\mu,\nu)$ is a transport maps from the Lebesgue measure to itself. Moreover, note that the induced coupling $\pi_n := (Id_{[0,1]}, T_n)_{\#}\mu$ converges weakly to

$$\pi := \frac{1}{2}(Id_{[0,1]}, Id_{[0,1]})_{\#}\mu + (Id_{[0,1]}, 1 - Id_{[0,1]})_{\#}\mu)$$

because half of the mass is sent to $y = x$ and the other half is sent to $y = 1-x$. However, π is not induced by any transport map, since its support in $[0,1]^2$ is not the graph of a function. Therefore π is a proper coupling between μ and ν, and $\pi \notin \mathcal{T}(\mu,\nu)$.

1.4 Kantorovich and Monge Problems

In this section, we finally the optimal transport problems, both in the Monge formulation and the Kantorovich formulation, and then we will study some properties of these problems.
The first notion we introduce models the cost of transporting one unity of mass from a point x of the domain space X to another point y in the arrival space Y.

Definition 1.4.1. A cost function for an optimal transport problem with distributions $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$ is a measurable map $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$.
In particular, we say that c is bounded below if

$$c(x,y) \geq a(x) + b(y) \text{ for all } x \in X, y \in Y,$$

for some upper-semicontinuous functions $a : X \to \mathbb{R} \cup \{-\infty\}$, $b : Y \to \mathbb{R} \cup \{-\infty\}$ with $a \in L^1(\mu)$, $b \in L^1(\nu)$.

In many practical examples, we will consider the setting $X = Y$ with cost given by the metric $d : X \times X \to [0, \infty)$ induced on the Polish space X.

Remark 1.4.2. The lower bound assumption $c(x, y) \geq a(x) + b(y)$ guarantees that the expected cost $\mathbb{E}[c(X, Y)]$ is well defined in $\mathbb{R} \cup \{+\infty\}$, with $X \sim \mu,\ Y \sim \nu$.
In most of the cases, we will consider non-negative cost functions $c : X \times Y \to [0, +\infty]$, i.e. with $a = 0, b = 0$.

Definition 1.4.3. The Monge functional with cost $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ is the map $P_c^M(\mu, \nu) : \mathcal{P}(X) \times \mathcal{P}(Y) \to \mathbb{R} \cup \{+\infty\}$ such that

$$P_c^M(\mu, \nu) := \inf_{T \in \mathcal{T}(\mu,\nu)} \int_X c(x, T(x))\ d\mu(x) \quad (MP)_c$$

In particular, a transport map $T \in \mathcal{T}(\mu, \nu)$ is called optimal transport if it attains the infimum of (MP).

Definition 1.4.4. The Kantorovich functional with cost $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ is the map $P_c^K(\mu, \nu) : \mathcal{P}(X) \times \mathcal{P}(Y) \to \mathbb{R} \cup \{+\infty\}$ such that

$$P_c^K(\mu, \nu) := \inf_{\pi \in Cpl(\mu,\nu)} \int_{X \times Y} c\ d\pi \quad (KP)_c$$

In particular, a coupling $\pi \in Cpl(\mu, \nu)$ is called optimal transport plan if it attains the infimum of $(KP)_c$.

Intuitively, both problems represent a minimization problem of an expected cost, depending on some marginal distributions μ, ν. Indeed, we can rewrite the functionals $P_c^M(\mu, \nu)$ and $P_c^K(\mu, \nu)$ as

$$P_c^M(\mu, \nu) := \inf_{\substack{T \in \mathcal{T}(\mu,\nu), \\ X \sim \mu}} \mathbb{E}[c(X, T(X))], \quad P_c^K(\mu, \nu) := \inf_{\substack{\pi \in Cpl(\mu,\nu) \\ X \sim \mu, Y \sim \nu}} \mathbb{E}_\pi[c(X, Y)].$$

Remark 1.4.5. The Kantorovich problem is a generalization of the Monge problem, since each transport map can be interpreted as a coupling of measures via canonical injection. However, (KP) is much better behaved than (MP), as we saw in the previous section:

- $Cpl(\mu, \nu)$ is non-empty, convex, and compact resp. weak topology.
- $\mathcal{T}(\mu, \nu)$ is in general not convex and not closed resp. weak topology

In particular, the convexity of (KP) endows the problem with a richer structure, allowing the use of a wide range of tools from convex analysis.

1.5 Existence

The first natural question arising from the definitions of (KP) and (MP) is the existence of optimizers, which will be clarified in the next theorem.

Definition 1.5.1. A function $f : X \to \bar{\mathbb{R}}$ is lower-semicontinuous (LSC) if

$$\liminf_{n \to \infty} f(x_n) \geq f(x)$$

for all sequences $\{x_n\}_n \subseteq X$ converging to $x \in X$.

Lemma 1.5.2. *Let $f : X \to \bar{\mathbb{R}}$ be a function, then the following hold:*

(i) f LSC $\iff \{f \leq y\} \subseteq X$ closed $\iff \{f > y\} \subseteq X$ open.

(ii) If f LSC, then f attains minimum on any $K \subseteq X$ compact.

(iii) (Baire Theorem) f LSC $\iff f = \lim_{n\to\infty} f_n$ for some $\{f_n\}_n \subseteq C(X),\ f_n \uparrow f$.

Lemma 1.5.3. *Let $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ be a lower-semicontinuous cost function such that $c \geq h$ for some upper-semicontinuous function $h : X \times Y \to \mathbb{R} \cup \{-\infty\}$. If $\{\pi_n\}_n$ sequence of probability measures in $\mathcal{P}(X \times Y)$ converging weakly to a probability π such that $h \in L^1(\pi_n),\ h \in L^1(\pi)$, and*

$$\int_{X\times Y} h \ d\pi_n \underset{n\to\infty}{\longrightarrow} \int_{X\times Y} h \ d\pi,$$

then

$$\int_{X\times Y} c \ d\pi \leq \liminf_{n\to\infty} \int_{X\times Y} c \ d\pi_n.$$

In particular, if $c \geq 0$ then the map

$$\pi \mapsto \int_{X\times Y} c \ d\pi$$

is lower-semicontinuous.

Proof. Note that we may assume that c is non-negative, otherwise we can replace c with $c - h \geq 0$. Then, c can be written as the pointwise limit of a non decreasing family $\{c_k\}_k$ of continuous and bounded functions by Lemma 1.5.2, i.e. $c_k \uparrow c$. Then, we have

$$\begin{aligned}\int_{X\times Y} c \ d\pi &= \lim_{k\to\infty} \int_{X\times Y} c_k \ d\pi = \text{ by monotone convergence} \\ &= \lim_{k\to\infty} \lim_{n\to\infty} \int_{X\times Y} c_k \ d\pi_n \text{ by weak convergence}\end{aligned}$$

Now, note that

$$\int_{X\times Y} c_k \ d\pi_n \leq \int_{X\times Y} c \ d\pi_n \text{ for all } n \in \mathbb{N}$$

since $\{c_n\}_n$ non decreasing sequence converging to c. Hence, by taking the liminf for $n \to \infty$ we get for all $k \in \mathbb{N}$

$$\int_{X\times Y} c_k \ d\pi = \lim_{n\to\infty} \int_{X\times Y} c_k \ d\pi_n = \liminf_{n\to\infty} \int_{X\times Y} c_k \ d\pi_n \leq \liminf_{n\to\infty} \int_{X\times Y} c \ d\pi_n,$$

by weak convergence of π_n to π. Finally, the thesis follows by taking the lim for $k \in \mathbb{N}$

$$\lim_{k\to\infty} \lim_{n\to\infty} \int_{X\times Y} c_k \ d\pi_n \leq \liminf_{n\to\infty} \int_{X\times Y} c \ d\pi_n.$$

$\square$

Theorem 1.5.4. *(Optimal Transport Plans Existence) Let $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ be a lower-semicontinuous and bounded below cost function for a Kantorovich problem with marginals $\mu \in \mathcal{P}(X),\ \nu \in \mathcal{P}(Y)$. Then, there exists an optimal transport plan for the Kantorovich problem $(KP)_c$, i.e.*

$$\exists\ \pi^* \in argmin_{\pi \in Cpl(\mu,\nu)} \int_{X \times Y} c\ d\pi.$$

Proof. Let $\{\pi_n\}_n \subseteq Cpl(\mu,\nu)$ be a sequence of couplings between μ and ν with expected cost converging to the infimum average cost, i.e. such that

$$\int_{X \times Y} c\ d\pi_n \underset{n\to\infty}{\longrightarrow} P_c^K(\mu,\nu) = \inf_{\pi \in Cpl(\mu,\nu)} \int_{X \times Y} c\ d\pi$$

Then, there exists a subsequence $\{\pi_{n_k}\}_k \subseteq Cpl(\mu,\nu)$ converging weakly to a coupling $\pi \in Cpl(\mu,\nu)$, since $Cpl(\mu,\nu)$ is compact by Proposition 1.3.7. Moreover, note that $c \geq h$ with $h(x,y) = a(x) + b(y)$, and it also holds $h \in L^1(\pi_n),\ h \in L^1(\pi)$ with

$$\int_{X \times Y} h\ d\pi_n = \int_{X \times Y} a\ d\mu + \int_{X \times Y} b\ d\nu = \int_{X \times Y} c\ d\pi$$

by Proposition 1.2.2, since $\pi_n, \pi \in Cpl(\mu,\nu)$.
Then, we get that π attains the infimum by Lemma 1.5.3 since the cost is LSC, i.e.

$$\int_{X \times Y} c\ d\pi \leq \liminf_{n\to\infty} \int_{X \times Y} c\ d\pi_n = P_c^K(\mu,\nu).$$

□

Remark 1.5.5. The existence theorem does not imply that the optimal cost is finite. It might be that all transport plans lead to an infinite total cost, i.e. $\int_{X \times Y} c\ d\pi = +\infty$ for all $\pi \in Cpl(\mu,\nu)$. A simple condition to rule out this annoying possibility is

$$\int_{X \times Y} c(x,y)\ d\mu(x)\ d\nu(y) < +\infty,$$

which guarantees that at least the independent coupling $\mu \otimes \nu$ has finite total cost. Alternatively, one can make the stronger assumption

$$c(x,y) \leq c_X(x) + c_Y(y) \text{ for some functions } c_X \in L^1(\mu),\ c_Y \in L^1(\nu).$$

1.6 Duality

The previous result guarantees the existence of optimizer for the Kantorovich problem with rather mild assumptions on the regularity of the cost function c. Once existence is established, some other natural questions arise:

- What is the dual formulation of the convex problem (KP)?
- Is the solution of (KP) unique?
- Do (MP) and (KP) have the same solutions?
- Is there any structure in the optimizer π^* of (KP)?

Definition 1.6.1. The dual Kantorovich functional with cost $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ is the map $D_c^K(\mu,\nu) : \mathcal{P}(X) \times \mathcal{P}(Y) \to \mathbb{R} \cup \{+\infty\}$ such that

$$D_c^K(\mu,\nu) := \sup_{(\varphi,\psi)\in D(c)} \left(\int_X \varphi \; d\mu \; + \; \int_Y \psi \; d\nu \right) \quad \text{(DP)}$$

with $D(c) := \{(\varphi,\psi) \in L^1(\mu) \times L^1(\nu) \mid c \geq \varphi \oplus \psi\}$ research space.

Intuitively, if the Kantorovich problem (KP) aims to minimize some average cost, then the dual problem (DP) aims to maximize the some profit.
In the following we will use two distinct notions of duality, and we will say that:

- Weak Duality holds if $D_c^K(\mu,\nu) \leq P_c^K(\mu,\nu)$
- Strong Duality holds if $D_c^K(\mu,\nu) = P_c^K(\mu,\nu)$

Proposition 1.6.2. *(Weak Duality) Let $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ be a cost function for a Kantorovich problem with marginals $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$. Then, weak duality holds, i.e. $D_c^K(\mu,\nu) \leq P_c^K(\mu,\nu)$.*

Proof. Let $\varphi \in L^1(\mu), \psi \in L^1(\nu)$ be integrable functions such that $\varphi \oplus \psi \leq c$, and consider a coupling $\pi \in Cpl(\mu,\nu)$. Then, we have

$$\begin{aligned} \int_X \varphi \; d\mu \; + \; \int_Y \psi \; d\nu &= \int_{X\times Y} (\varphi \oplus \psi) \; d\pi \text{ by Proposition 1.2.2} \\ &\leq \int_{X\times Y} c \; d\pi. \end{aligned}$$

Hence, the thesis follows by taking the infimum of RHS over $\pi \in Cpl(\mu,\nu)$, and the supremum of the LHS over $(\varphi,\psi) \in D(c)$, that is

$$D_c^K(\mu,\nu) = \sup_{(\varphi,\psi)\in D(c)} \left(\int_X \varphi \; d\mu \; + \; \int_Y \psi \; d\nu \right) \leq \inf_{\pi\in Cpl(\mu,\nu)} \int_{X\times Y} c \; d\pi = P_c^K(\mu,\nu)$$

□

Remark 1.6.3. Strong duality may not hold.
For example, consider $X = Y = [0,1]$ with (non lower-semicontinuous) cost

$$c(x,y) = \begin{cases} 0 & x < y \\ 1 & x = y \\ +\infty & \text{otherwise.} \end{cases}$$

Then, duality does not hold for the Kantorovich problem with marginals $\mu = \nu = \lambda_{|[0,1]}$. Indeed, the identity coupling $\pi := (Id_{[0,1]}, Id_{[0,1]})\lambda \in Cpl(\lambda,\lambda)$ realizes the infimum of the Kantorovich problem

$$\inf_{\pi\in Cpl(\lambda,\lambda)} \int_{[0,1]^2} c \; d\pi = 1,$$

since π supported on the diagonal of $[0,1]^2$. However, the dual problem satisfies

$$\sup_{\varphi\oplus\psi\leq c} \int_{[0,1]} \varphi(x) + \psi(x) \; dx = 0.$$

In order to establish a theorem for the Strong Duality, we first need the following technical result.

Lemma 1.6.4. *(Min-Max Principle) Let H_1, H_2 be vector spaces with H_1 locally compact, and let $K \subseteq H_1$ be a convex and compact subspace, $L \subseteq H_2$ be a convex subspace. Let $F : K \times L \to \mathbb{R}$ be a map such that*

(i) $x \mapsto F(x, y)$ continuous and convex for all $y \in L$

(ii) $y \mapsto F(x, y)$ concave for all $x \in K$

Then,

$$\sup_{y \in L} \inf_{x \in K} F(x, y) = \inf_{x \in K} \sup_{y \in L} F(x, y)$$

Proof. Theorem 37.1, [6]. $\square$

Theorem 1.6.5. *(Duality, I) Let $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ be a lower-semicontinuous and bounded below cost function for a Kantorovich problem with marginals $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$. Then, the following hold:*

(i) Strong duality holds, i.e.

$$D_c^K(\mu, \nu) = P_c^K(\mu, \nu).$$

(ii) The research space $D(c)$ of the dual problem can be restricted to

$$\tilde{D}(c) := \{(\varphi, \psi) \in C_b(X) \times C_b(Y) \mid c \geq \varphi \oplus \psi\},$$

i.e., it holds

$$\sup_{\substack{(\varphi,\psi) \in L^1(\mu) \times L^1(\nu) \\ c \geq \varphi \oplus \psi}} \left(\int_X \varphi \, d\mu \; + \; \int_Y \psi \, d\nu \right) = \sup_{\substack{(\varphi,\psi) \in C_b(X) \times C_b(Y) \\ c \geq \varphi \oplus \psi}} \left(\int_X \varphi \, d\mu \; + \; \int_Y \psi \, d\nu \right).$$

(iii) Let $\varphi^* \in L^1(\mu), \psi^* \in L^1(\nu)$ be optimizer of the dual problem, then

$$\pi \in Cpl(\mu, \nu) \text{ optimal} \iff c = \varphi^* \oplus \psi^* \quad \pi - a.s.$$

Similarly, if π^* is an optimizer of the Kantorovich problem, then

$$\varphi \in L^1(\mu), \ \psi \in L^1(\nu) \text{ s.t. } \varphi \oplus \psi \leq c \text{ optimal} \iff c = \varphi \oplus \psi \quad \pi^* - a.s.$$

Proof.
(i) We will prove the result in the particular case where X, Y are compact spaces and the cost $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ is a continuous function.
Let μ, ν be probability measures on X, Y respectively, and we prove that $P_c^K(\mu, \nu) = D_c^K(\mu, \nu)$. Consider the map $\chi : \mathcal{P}(X \times Y) \to \{0, \infty\}$ such that

$$\chi(\pi) := \begin{cases} 0 & \pi \in Cpl(\mu, \nu) \\ +\infty & \text{otherwise} \end{cases}$$

Note that for a probability measure $\pi \in \mathcal{P}(X \times Y)$ we have by Proposition 1.2.2

$$\pi \in Cpl(\mu, \nu) \iff \int_{X \times Y} \varphi \oplus \psi \, d\pi = \int_X \varphi \, d\mu + \int_Y \psi \, d\nu.$$

Then, we can rewrite the map $\pi \to \chi(\pi)$ as

$$\chi(\pi) = \sup_{(\varphi,\psi)\in L^1(\mu)\times L^1(\nu)} \left(\int_X \varphi \, d\mu + \int_Y \psi \, d\nu - \int_{X\times Y} \varphi \oplus \psi \, d\pi \right).$$

Therefore, that Kantorovich problem is given by

$$\begin{aligned} P_c^K(\mu,\nu) &= \inf_{\pi\in Cpl(\mu,\nu)} \int_{X\times Y} c \, d\pi = \inf_{\pi\in\mathcal{P}(X\times Y)} \left(\chi(\pi) + \int_{X\times Y} c \, d\pi \right) = \\ &= \inf_{\pi\in\mathcal{P}(X\times Y)} \sup_{(\varphi,\psi)\in L^1(\mu)\times L^1(\nu)} \left(\int_X \varphi \, d\mu + \int_Y \psi \, d\nu + \int_{X\times Y} (c - \varphi\oplus\psi) \, d\pi \right) \end{aligned}$$

Now, note that the space $\mathcal{P}(X\times Y)$ is compact for X, Y compact spaces, and we can apply the min-max principle with $K = \mathcal{P}(X\times Y)$ and $L = L^1(\mu)\times L^1(\nu)$. Thus, we have

$$\begin{aligned} P_c^K(\mu,\nu) &= \inf_{\pi\in\mathcal{P}(X\times Y)} \sup_{(\varphi,\psi)\in L^1(\mu)\times L^1(\nu)} \left(\int_X \varphi \, d\mu + \int_Y \psi \, d\nu + \int_{X\times Y} (c - \varphi\oplus\psi) \, d\pi \right) \\ &= \sup_{(\varphi,\psi)\in L^1(\mu)\times L^1(\nu)} \inf_{\pi\in\mathcal{P}(X\times Y)} \left(\int_X \varphi \, d\mu + \int_Y \psi \, d\nu + \int_{X\times Y} (c - \varphi\oplus\psi) \, d\pi \right) \\ &\le \sup_{(\varphi,\psi)\in L^1(\mu)\times L^1(\nu)} \inf_{(x,y)\in X\times Y} \left(\int_X \varphi \, d\mu + \int_Y \psi \, d\nu + c(x,y) - \varphi(x) - \psi(y) \right) \\ &= \sup_{\substack{(\varphi,\psi)\in L^1(\mu)\times L^1(\nu) \\ c\ge\varphi\oplus\psi}} \left(\int_X \varphi \, d\mu + \int_Y \psi \, d\nu \right) = D_c^K(\mu,\nu). \end{aligned}$$

More generally, when X, Y are not compact and the cost c is not continuous, we can approximate the cost function c with a non-decreasing sequence of continuous functions $\{c_n\}_n$, and we can approximate μ, ν with compactly supported probability measures $\tilde{\mu}, \tilde{\nu}$, and use the previous step.

(ii) Note that by density of $C_b(X) \subseteq L^1(\mu)$ and $C_b(Y) \subseteq L^1(\nu)$, the research space of maximizers can be restricted to the space

$$\tilde{D}(c) := \{(\varphi,\psi) \in C_b(X)\times C_b(Y) \mid c \ge \varphi\oplus\psi\}.$$

(iii) Let $\varphi^* \in L^1(\mu)$, $\psi^* \in L^1(\nu)$ be optimizer of the dual problem, and assume that $\pi \in Cpl(\mu,\nu)$ satisfies $c = \varphi^*\oplus\psi^*$ π-a.s. Then, we have

$$\begin{aligned} \int_{X\times Y} c \, d\pi = \int_{X\times Y} \varphi^*\oplus\psi^* \, d\pi &= \int_X \varphi^* \, d\mu + \int_Y \psi^* \, d\nu \\ &= \sup_{\substack{(\varphi,\psi)\in L^1(\mu)\times L^1(\nu) \\ c\ge\varphi\oplus\psi}} \left(\int_X \varphi \, d\mu \; + \; \int_Y \psi \, d\nu \right) \\ &= \inf_{\pi\in Cpl(\mu,\nu)} \int_{X\times Y} c \, d\pi \quad \text{by duality.} \end{aligned}$$

Hence, π attains the infimum of the Kantorovich problem, and is then an optimizer. $\square$

The next relevant result originates from a simple question:

Given $(\varphi, \psi) \in L^1(\mu) \times L^1(\nu)$ such that $\varphi \oplus \psi \leq c$, achieving a certain score $\int_X \varphi \, d\mu + \int_X \psi \, d\nu$ in the dual problem, can we construct a couple $(\tilde{\varphi}, \tilde{\psi})$ still satisfying $\tilde{\varphi} \oplus \tilde{\psi} \leq c$ and achieving a better core, i.e. such that

$$\int \varphi \, d\mu + \int \psi \, d\nu \leq \int \tilde{\varphi} \, d\mu + \int \tilde{\psi} \, d\nu?$$

For example, one may fix the function φ, and improve ψ by considering the function

$$\tilde{\psi}(y) := \inf_{x \in X} \, (c(x, y) - \varphi(x)).$$

Then, we have

$$\varphi(x) + \tilde{\psi}(y) = \varphi(x) + \inf_{\bar{x} \in X} \, (c(\bar{x}, y) - \varphi(\bar{x})) \leq \varphi(x) + c(x, y) - \varphi(x) = c(x, y)$$

and

$$\int \varphi \, d\mu + \int \psi \, d\nu \leq \int \varphi \, d\nu + \int \tilde{\psi} \, d\nu$$

since $\psi \leq \tilde{\psi}$ by $\varphi \oplus \psi \leq c$.

Definition 1.6.6. The transform of a measurable function $\varphi : X \to \mathbb{R}$ respect to a cost function $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$, or c-transform of φ, is the map $\varphi^c : Y \to \mathbb{R}$ such that

$$\varphi^c(y) := \inf_{x \in X} (c(x, y) - \varphi(x)).$$

Similarly, the c-transform of a map $\psi : Y \to \mathbb{R}$ is the function $\psi^c : X \to \mathbb{R}$ such that

$$\psi^c(x) := \inf_{y \in Y} (c(x, y) - \psi(y)).$$

Moreover, we say that φ is c-concave is $\varphi = \psi^c$ for some map ψ, and similarly ψ is c-concave if $\psi = \varphi^c$ for some function φ.

Example 1.6.7.

(i) *Quadratic Cost* $c(x, y) = -x \cdot y$
Let $X = Y = \mathbb{R}^d$ with cost $c(x, y) = -x \cdot y$, and let $\varphi : \mathbb{R}^d \to \mathbb{R}$ be a measurable function. Then,

$$\varphi : \mathbb{R}^d \to \mathbb{R} \text{ } c\text{-concave} \iff \varphi \text{ concave and USC}$$

In particular, the c-transform is given by

$$\varphi^c(y) = \inf_{x \in \mathbb{R}^d} \, (-x \cdot y - \varphi(x)) = -\sup_{x \in \mathbb{R}^d} \, (x \cdot y - (-\varphi)(x)) = -(-\varphi)^*(y),$$

where $\varphi^* : \mathbb{R}^d \to \mathbb{R}$ Legendre transform of φ.
Indeed, suppose that $\varphi = \psi^c$ for some $\psi : \mathbb{R}^d \to \mathbb{R}$, then we have $\varphi = -(-\psi)^*$ concave and upper-semicontinuous, since the Legendre transform is convex and lower-semicontinuous. Conversely, if φ is concave and upper-semicontinuous, then we can define $\psi := -(-\varphi)^*$ and obtain

$$\psi^c = -(-\psi)^* = -(-\varphi)^{**} = \varphi$$

by Fenchel-Moreau theorem, since $-\varphi$ is convex and lower-semicontinuous.

(ii) *Metric Cost* $c(x,y) = d(x,y)$
Let $X = Y$ with $c(x,y) = d(x,y)$, and let $\varphi : X \to \mathbb{R}$ be a measurable function. Then,

$$\varphi : X \to \mathbb{R} \ c\text{-concave} \iff \varphi \ \text{1-Lipschitz, with } \varphi^c = -\varphi$$

Indeed, if $\varphi = \psi^c$ for some $\psi : X \to \mathbb{R}$ measurable, then we have

$$\begin{aligned}|\varphi(x_1) - \varphi(x_2)| &= |\inf_{y \in X} (d(x_1,y) - \psi(y)) - \inf_{y\in X} (d(x_2,y) - \psi(y))| \\ &\leq |\inf_{y\in X} (d(x_1,y) - \psi(y) - d(x_2,y) + \psi(y))| \\ &\leq |\inf_{y\in X} d(x_1,x_2)| = d(x_1,x_2)\end{aligned}$$

by triangle inequality, and φ is 1-Lipschitz.
Viceversa, suppose that φ is 1-Lipschitz, define $\psi := -\varphi$ and obtain

$$\psi^c(x) = \inf_{y\in X} (d(x,y) + \varphi(y)) \leq d(x,x) + \varphi(x) = \varphi(x)$$

and

$$\varphi(x) \leq \inf_{y\in X} (d(x,y) + \varphi(y)) = \inf_{y\in X} (d(x,y) - \psi(y)) = \psi^c(x) \quad \text{by Lipschitz.}$$

In particular, the computation also shows that the c-transform of φ is $\varphi^c = -\varphi$.

(iii) *Quadratic Cost* $c(x,y) = \frac{1}{2}|x-y|^2$
Let $X = Y = \mathbb{R}^d$ with $c(x,y) = \frac{1}{2}|x-y|^2$, and let $\varphi : \mathbb{R}^d \to \mathbb{R}$ be a measurable function. Then,

$$\varphi : \mathbb{R}^d \to \mathbb{R} \ c\text{-concave} \iff \frac{|x|^2}{2} - \varphi(x) \ \text{convex and LSC.}$$

Note that the cost function $c(x,y) = \frac{1}{2}|x-y|^2$ has essentially the same structure of the cost $-x \cdot y$, since the interaction term between x, y is the same. Indeed, the c-transform is given by

$$\begin{aligned}\varphi^c(y) &= \inf_{x\in\mathbb{R}^d} \left(\frac{1}{2}|x-y|^2 - \varphi(x)\right) = \\ &= \inf_{x\in\mathbb{R}^d} \left(\frac{1}{2}|x|^2 - x\cdot y + \frac{1}{2}|y|^2 - \varphi(x)\right) \\ &= \frac{1}{2}|y|^2 + \inf_{x\in\mathbb{R}^d} \left(-x\cdot y + \frac{1}{2}|x|^2 - \varphi(x)\right) \\ &= \frac{1}{2}|y^2| + \left(\frac{1}{2}|\cdot|^2 - \varphi\right)^* (y)\end{aligned}$$

Then, one can prove similarly to (i), that φ is c-concave if and only if $\frac{1}{2}|\cdot|^2 - \varphi$ is convex and lower-semicontinuous, by Fenchel-Moreau theorem.

The previous example shows that we can interpret c-transforms as a generalization of the Legendre transform in $\mathbb{R}^d$ with quadratic cost $c(x,y) = -x\cdot y$. Now, we recall some results about the Legendre transform of a function and its biconjugate.

Proposition 1.6.8. *Let $\varphi : \mathbb{R}^d \to \mathbb{R}$ be a measurable function. Then, the Legendre transform φ^* is convex and lower-semicontinuous.*

Theorem 1.6.9. *(Fenchel-Moreau) Let $\varphi : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be a function. Then, the biconjugate φ^{**} is the largest lower-semicontinuous convex function with $\varphi^{**} \leq \varphi$.*
In particular, $\varphi^{**} = \varphi$ if and only if φ convex and lower-semicontinuous.

Proof. Theorem 12.2, [6]. □

The following result generalizes the properties of the Legendre transform φ^* and the biconjugate φ^{**} to the case of c-transforms.

Proposition 1.6.10. *Let $c : X \times Y \to \mathbb{R}$ be a cost function, and $\varphi : X \to \mathbb{R}$ measurable. Then, it holds:*

(i) φ^c c-concave, upper-semicontinuous for c continuous.

(ii) $\varphi^{cc} \geq \varphi$, with equality if and only if φ c-concave. Moreover, φ^{cc} is smallest c-concave function larger than φ, called c-concavification of φ.

Proof.
(i) Note that if the cost function c is continuous, then the map $y \mapsto c(x,y) - \varphi(x)$ is continuous for all $x \in X$. Therefore, the map

$$\varphi^c(y) = \inf_{x \in X} (c(x,y) - \varphi(y))$$

is upper-semicontinuous, as a infimum of continuous functions.
(ii) By definition of c-transform we have

$$\begin{aligned}
\varphi^{cc}(x) &= \inf_{y \in Y} (c(x,y) - \varphi^c(y)) \\
&= \inf_{y \in Y} (c(x,y) - \inf_{\tilde{x} \in X} (c(\tilde{x},y) - \varphi(\tilde{x}))) \\
&= \inf_{y \in Y} \underbrace{\sup_{\tilde{x} \in X}(c(x,y) - c(\tilde{x},y) + \varphi(\tilde{x}))}_{\geq c(x,y) - c(x,y) + \varphi(x) = \varphi(x)} \\
&\geq \inf_{y \in Y} \varphi(x) = \varphi(x)
\end{aligned}$$

Now, note that $\varphi^{ccc} = \varphi$, since

$$\begin{aligned}
\varphi^{ccc}(y) &= \inf_{x \in X} (c(x,y) - \varphi^{cc}(x)) \\
&= \inf_{x \in X} (c(x,y) - \inf_{\tilde{y} \in Y} \sup_{\tilde{x} \in X}(c(x,\tilde{y}) - c(\tilde{x},\tilde{y}) + \varphi(\tilde{x}))) \\
&= \inf_{x \in X} \sup_{\tilde{y} \in Y} \inf_{\tilde{x} \in X} (c(x,y) - c(x,\tilde{y}) + c(\tilde{x},\tilde{y}) - \varphi(\tilde{x}))
\end{aligned}$$

Then, by choosing $\tilde{y} = y$ we get

$$\begin{aligned}
\varphi^{ccc}(y) &= \inf_{x \in X} \sup_{\tilde{y} \in Y} \inf_{\tilde{x} \in X} (c(x,y) - c(x,\tilde{y}) + c(\tilde{x},\tilde{y}) - \varphi(\tilde{x})) \\
&\geq \inf_{x \in X} \inf_{\tilde{x} \in X} (c(x,y) - c(x,y) + c(\tilde{x},y) - \varphi(\tilde{x})) \\
&= \inf_{\tilde{x} \in X} (c(\tilde{x},y) - \varphi(\tilde{x})) = \varphi^c(y)
\end{aligned}$$

Conversely, by choosing $\tilde{x} = x$ we obtain $\varphi^{ccc} \leq \varphi$, and conclude that $\varphi^{ccc} = \varphi$.
Note that φ^{cc} is c-concave by definition. Viceversa, if φ c-concave, then $\varphi = \psi^c$ for some $\psi : Y \to \mathbb{R}$, and we have $\varphi^{cc} = (\psi^c)^{cc} = \psi^c = \varphi$ by previous computation. □

Finally, we state the duality theorem in its most general form.

Theorem 1.6.11. *(Duality, II) Let $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ be a lower-semicontinuous and bounded below cost function for a Kantorovich problem with marginals $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$. Then,*

$$\begin{aligned}\inf_{\pi \in Cpl(\mu,\nu)} \int_{X\times Y} c \, d\pi &= \sup_{\substack{(\varphi,\psi)\in C_b(X)\times C_b(Y)\\ c\geq\varphi\oplus\psi}} \left(\int_X \varphi \, d\mu \; + \; \int_Y \psi \, d\nu \right) \\ &= \sup_{\substack{(\varphi,\psi)\in L^1(\mu)\times L^1(\nu)\\ c\geq\varphi\oplus\psi}} \left(\int_X \varphi \, d\mu \; + \; \int_Y \psi \, d\nu \right) \\ &= \sup_{\psi\in L^1(\nu)} \left(\int_X \psi^c \, d\mu + \int_Y \psi \, d\nu \right) \\ &= \sup_{\varphi\in L^1(\mu)} \left(\int_X \varphi \, d\mu + \int_Y \varphi^c \, d\nu \right)\end{aligned}$$

Moreover, if $\varphi^* \in L^1(\mu), \psi^* \in L^1(\nu)$ be optimizer of the dual problem, then

$$\pi \in Cpl(\mu,\nu) \textit{ optimal} \iff c = \varphi^* \oplus \psi^* \quad \pi - a.s.$$

Similarly, if π^* is an optimizer of the Kantorovich problem, then

$$\varphi \in L^1(\mu), \;\; \psi \in L^1(\nu) \textit{ s.t. } \varphi \oplus \psi \leq c \textit{ optimal} \iff c = \varphi \oplus \psi \quad \pi^* - a.s.$$

Proof. The thesis follows by Theorem 1.6.5, by noting that for any $\varphi : X \to \mathbb{R}$, $\psi : Y \to \mathbb{R}$ it holds $\varphi + \varphi^c \leq c$ by definition of c-transform, with

$$\int_X \varphi \, d\mu + \int_Y \psi \, d\nu \leq \int_X \varphi \, d\mu + \int_Y \varphi^c \, d\nu.$$

□

A particular case of the duality theorem is the Kantorovich-Rubenstein formula.

Corollary 1.6.12. *(Kantorovich-Rubenstein) The Kantorovich problem with metric cost $c(x,y) = d(x,y)$ on a Polish space X, and marginals $\mu, \nu \in \mathcal{P}_1(X)$ satisfies the duality*

$$\inf_{\pi\in Cpl(\mu,\nu)} \int_{X\times Y} c \, d\pi = \sup_{\varphi \;\, 1-Lip} \left(\int_X \varphi \, d\mu - \int_X \varphi \, d\nu \right).$$

Moreover, the problem both problems admit optimizers. In particular, if φ^* 1-Lipschitz optimizer, then

$$\pi \in Cpl(\mu,\nu) \textit{ optimal} \iff \varphi^*(x) - \varphi^*(y) = d(x,y) \textit{ for } \pi - a.s. \; x, y \in X.$$

Proof. Note that the cost function $c(x,y) = d(x,y)$ is continuous, then we get by duality theorem

$$\begin{aligned}\inf_{\pi\in Cpl(\mu,\nu)} \int c \, d\pi &= \sup_{\varphi\in L^1(\mu)} \left(\int_X \varphi \, d\nu + \int_Y \varphi^c \, d\nu \right) \\ &= \sup_{\varphi \;\, 1-\mathrm{Lip}} \left(\int_X \varphi \, d\nu - \int_Y \varphi \, d\nu \right)\end{aligned}$$

since the transform of a measurable map φ is $\varphi^c = -\varphi$, and the c-concave functions are exactly the Lipschitz functions of constant 1. □

1.7 Structural Results

Now, we want to better study the structure of the space of optimizers of the Kantorovich problem.

Definition 1.7.1. A set $\Gamma \subseteq X \times Y$ is cyclically monotone respect to a cost function $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$, or also c-cyclically monotone ($c - cm$), if

$$\sum_{i=1}^{n} c(x_i, y_i) \leq \sum_{i=1}^{n} c(x_i, y_{i+1})$$

for all finite sequences $(x_1, y_1), \dots, (x_n, y_n) \in \Gamma$.
Equivalently, $\Gamma \subseteq X \times Y$ is c-cyclically monotone if

$$\sum_{i=1}^{n} c(x_i, y_i) \leq \sum_{i=1}^{n} c(x_i, y_{P(i)})$$

for all sequences $(x_1, y_1), \dots, (x_n, y_n) \in \Gamma$, and all indices permutations $P \in Sym(n)$.

Remark 1.7.2. The cyclically monotonicity is a geometric condition on the support of couplings, and is independent of the distribution of mass given by the coupling.

Definition 1.7.3. A probability measure $\pi \in \mathcal{P}(X \times Y)$ is c-cyclically monotone for a cost $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ if $\pi(\Gamma) = 1$ for some c-cyclically monotone set $\Gamma \subseteq X \times Y$.

Remark 1.7.4. In particular, if the support $supp(\pi) \subseteq X \times Y$ of π is c-cyclically monotone, then π is c-cyclically monotone. We will see in the following, that the geometric structure of $supp(\pi)$ is determinant for the optimality of the coupling.

Proposition 1.7.5. *Let $c : X \times Y \to [0, \infty]$ be a non-negative and continuous and bounded below cost function for a Kantorovich problem with marginals $\mu \in \mathcal{P}(X), \nu \in \mathcal{P}(Y)$. If $\pi^* \in Cpl(\mu, \nu)$ optimal coupling with finite value, then $supp(\pi) \subseteq X \times Y$ c-cyclically monotone.*

Proof. Assume by contradiction that

$$\sum_{i=1}^{n} c(x_i, y_i) > \sum_{i=1}^{n} c(x_i, y_{i+1})$$

for some $(x_1, y_1), \dots, (x_n, y_n) \in supp(\pi)$. Then, there exists $U_1, \dots, U_n$ open disjoint neighborhoods of $x_1, \dots, x_n$, and $V_1, \dots, V_n$ open disjoint neighborhoods of $y_1, \dots, y_n$ such that

$$\sum_{i=1}^{n} c(u_i, v_i) > \sum_{i=1}^{n} c(u_i, v_{i+1}) \tag{2}$$

for all $u_i \in U_i, v_i \in V_i$, by continuity of c. Now, consider the probability measures

$$\pi_i := \frac{1}{m_i} \pi^*_{|U_i \times V_i} \quad \text{for } i = 1, \dots, n,$$

where $\pi_{|U_i \times V_i}(A) = \pi^*(A \cap (U_i \times V_i))$ and $m_i = \pi^*(U_i \times V_i)$.
Finally, let μ_i, ν_i be the marginals of π_i, and consider the probability measure

$$\tilde{\pi} := \pi^* + \frac{\min_{i=1,\dots,n} m_i}{n} \left(\sum_{i=1}^{n} \mu_i \otimes \nu_{i+1} \; - \; \pi_i \right).$$

Note that

$$\pi^* - \frac{\min_{i=1,\dots,n} m_i}{n} \sum_{i=1}^{n} \pi_i \in Cpl\left(\mu - \frac{\min_{i=1,\dots,n} m_i}{n} \sum_{i=1}^{n} \mu_i,\ \nu - \frac{\min_{i=1,\dots,n} m_i}{n} \sum_{i=1}^{n} \nu_i\right)$$

by Lemma 1.2.6. Then, we have

$$\tilde{\pi} = \pi^* - \frac{\min_{i=1,\dots,n} m_i}{n} \sum_{i=1}^{n} \pi_i \ + \ \frac{\min_{i=1,\dots,n} m_i}{n} \sum_{i=1}^{n} \mu_i \otimes \nu_{i+1} \in Cpl(\mu, \nu).$$

Finally, one can prove that the coupling $\tilde{\pi} \in Cpl(\mu, \nu)$ achieves a lower average cost than the optimal coupling $\pi \in Cpl(\mu, \nu)$ using 2. $\square$

Now, we introduce another geometric notion which will be related to the structure of the support of optimal couplings.

Definition 1.7.6. The c-superdifferential of a c-concave function $\varphi : X \to \mathbb{R}$ is the set

$$\partial^c \varphi := \{(x, y) \in X \times Y \mid \varphi(x) + \varphi^c(y) = c(x, y)\}.$$

In particular, the c-superdifferential of φ at $x \in X$ is the set

$$\partial^c \varphi(x) = \{y \in Y \mid (x, y) \in \partial^c \varphi\}.$$

Remark 1.7.7. Intuitively, the c-superdifferential of a c-concave function φ contains the points satisfying the optimality of the dual problem.
Indeed, we have see in the duality theorem that, if φ optimizer of the dual problem, then a coupling $\pi \in Cpl(\mu, \nu)$ is optimal if and only if

$$1 = \pi(\{(x, y) \in X \times Y | \varphi(x) + \varphi^c(y) = c(x, y)\}) = \pi(\partial^c \varphi).$$

In particular, if any coupling π concentrated on the c-superdifferential $\partial^c \varphi$ of a c-concave function φ is optimal for the Kantorovich problem.

Theorem 1.7.8. *(Rockafellar) Let $c : X \times Y \to \mathbb{R}$ be a cost function, then*

$$\Gamma \subseteq X \times Y \ c - \textit{cyclically monotone} \iff \Gamma \subseteq \partial^c \varphi \textit{ for some c-concave function } \varphi$$

Proof. Theorem 24.8, [6]. $\square$

Theorem 1.7.9. *(Fundamental OT Theorem) Let $c : X \times Y \to \mathbb{R}$ be a lower-semicontinuous cost function such that $|c(x, y)| \leq a(x) + b(y)$ for some $a \in L^1(\mu), b \in L^1(\nu)$, and let $\pi \in Cpl(\mu, \nu)$ be a coupling of $\mu \in \mathcal{P}(X), \nu \in \mathcal{P}(Y)$ with finite value.*
Then, the following are equivalent:

(i) π optimizer of the Kantorovich problem $P_c^K(\mu, \nu)$.

(ii) supp(π) $\subseteq X \times Y$ c-cyclically monotone.

(iii) supp(π) $\subseteq \partial^c \varphi$ for some c-concave function φ.

Moreover, duality holds and $\varphi \in L^1(\mu), \varphi^c \in L^1(\nu)$ are the optimizers of the dual problem.

Proof. Firstly, we assume that the cost $c : X \times Y \to \mathbb{R}$ is a non-negative continuous function; otherwise we can replace $c(x, y)$ by $\tilde{c}(x, y) = c(x, y) + a(x) + b(y)$.
Then, note that the implications *(i)* $\implies$ *(ii)* and *(ii)* $\implies$ *(iii)* are guaranteed by Lemma 1.7.5, and Theorem 1.7.8 respectively.
Now, suppose that $supp(\pi) \subseteq \partial^c\varphi$ for some c-concave function φ, and we prove that the measure π is an optimal coupling. Let $\psi := \varphi^c$ be the c-transform of φ, and note that by definition of c-transform we have

$$|\varphi \oplus \psi| = |\varphi \oplus \varphi^c| \leq |c| \leq a \oplus b.$$

Then, $\varphi_+ \in L^1(\mu),\ \psi_+ \in L^1(\nu)$ since $a \in L^1(\mu),\ b \in L^1(\nu)$, and therefore the integrals $\int_X \varphi \, d\mu, \int_Y \psi \, d\nu$ are well-defined.
Now, consider the increasing sequences $\varphi_n := (\varphi \vee (-n)) \wedge n,\ \psi_n := (\psi \vee (-n)) \wedge n$, and note that

$$\varphi_n \to \varphi,\ \psi_n \to \psi \quad \text{for } n \to \infty.$$

Moreover, $\varphi_n \oplus \psi_n \leq c$ and hence $\varphi_n \in L^1(\mu),\ \psi_n \in L^1(\nu)$, since φ_n, ψ_n bounded in $[-n, n]$. Then, we have

$$\begin{aligned}
\int_{X\times Y} c \, d\pi &= \int_{X\times Y} \varphi \oplus \psi \, d\pi \quad \text{since } supp(\pi) \subseteq \partial^c\varphi \\
&= \lim_{n\to\infty} \int_{X\times Y} \varphi_n \oplus \psi_n \, d\pi \quad \text{by monotone convergence} \\
&= \lim_{n\to\infty} \int_X \varphi_n \, d\mu + \int_Y \psi_n \, d\nu \quad \text{since } \varphi_n \in L^1(\mu), \psi_n \in L^1(\nu) \\
&= \int_X \varphi \, d\mu + \int_Y \psi \, d\nu.
\end{aligned}$$

Thus, $\varphi \in L^1(\mu),\ \psi \in L^1(\nu)$ are optimizers of the dual problem, and the coupling π is an optimizer of the Kantorovich problem.
More generally, we can approximate the cost function c with a non-decreasing sequence of continuous functions and prove the result similarly. □

Remark 1.7.10. The fundamental theorem of optimal transport shows that the optimality of a coupling depends only on the geometry of the support relatively to the cost function, and does not depend on the distribution of mass on the support. For instance, if π is an optimal coupling, and $\tilde{\pi}$ is another coupling with $supp(\tilde{\pi}) \subseteq supp(\pi)$, then $\tilde{\pi}$ is also optimal, because the c-cyclical monotonicity is also inherited by subsets.

Remark 1.7.11. A particular case is realized when the c-superdifferential is

$$\partial^c\varphi(x) = \{*\} \quad \text{for } \mu - a.e.\ x \in X.$$

In this case, the optimal couplings $\pi \in Cpl(\mu, \nu)$ have their support $supp(\pi)$ contained in a space $\partial^c\varphi$ with only one degree of freedom. In other words, optimal couplings are supported on the graph of a function, i.e. the coupling is induced by a transport map.

Now, we will analyze some results about special cases of the classical optimal transport problem, namely the real case $X = Y = \mathbb{R}^d$ with quadratic cost.

Definition 1.7.12. A function $T : \mathbb{R}^d \to \mathbb{R}^d$ is monotone respect to a probability measure $\mu \in \mathcal{P}(\mathbb{R}^d)$ if $T = \nabla\varphi$ μ-a.s. for some convex function $\varphi : \mathbb{R}^d \to \mathbb{R}$.

Theorem 1.7.13. *(Rademacher) Let $f : \mathbb{R}^d \to \mathbb{R}$ be a convex function. Then, f is a.e. differentiable.*

Theorem 1.7.14. *The Kantorovich problem with quadratic cost $c(x, y) = x \cdot y$ on $\mathbb{R}^d$, and marginals $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ s.t. $\mu \ll \lambda$ admits a unique optimal coupling $\pi^* \in Cpl(\mu, \nu)$ of the form $\pi^* = (Id, \nabla\varphi)\mu$ for some convex function $\varphi : \mathbb{R}^d \to \mathbb{R}$. In particular, there exists a unique optimal transport map $T : \mathbb{R}^n \to \mathbb{R}^n$ from μ to ν of the form $T = \nabla\varphi$ for some convex function $\varphi : \mathbb{R}^n \to \mathbb{R}$.*

Proof. Note that the existence of an optimizer is guaranteed by the continuity of the quadratic cost $c(x, y) = x \cdot y$, by Theorem 1.5.4. Moreover, c is bounded as

$$|c(x, y)| = |x \cdot y| \leq \frac{1}{2}|x|^2 + \frac{1}{2}|y|^2$$

with $a(x) = \frac{1}{2}|x|^2 \in L^1(\mu)$, $b(y) = \frac{1}{2}|y|^2 \in L^1(\nu)$, since μ absolutely continuous respect to the Lebesgue measure. Then, a coupling $\pi \in Cpl(\mu, \nu)$ is optimal if and only if $supp(\pi) \subseteq \partial^c\varphi$ for some $\varphi : \mathbb{R}^d \to \mathbb{R}$ c-concave function, by Theorem 1.7.9.
Recall that, for the quadratic cost $c(x, y) = x \cdot y$, a measurable map $\varphi : \mathbb{R}^d \to \mathbb{R}$ we have

$$\varphi \text{ } c\text{-concave} \iff \varphi \text{ convex and LSC}$$

Therefore, a coupling $\pi \in Cpl(\mu, \nu)$ is optimal if and only if $supp(\pi) \subseteq \partial^c\varphi$ for some $\varphi : \mathbb{R}^d \to \mathbb{R}$ convex and lower-semicontinuous. In particular, φ is μ-a.e. differentiable by Rademacher theorem, and the gradient $\nabla\varphi$ is μ-a.e. well-defined on $\mathbb{R}^d$, and consequently

$$\nabla\varphi(x) = y \iff y \in \partial^c\varphi(x) \quad \text{for } \mu - a.e.\ x \in \mathbb{R}^d.$$

Thus, the c-superdifferential is a singleton μ-a.e, and any optimal coupling π is supported on the graph of the gradient of differentiable functions.
Now, suppose that π_1, π_2 are optimal couplings supported on the graph of T_1, T_2 respectively. Then, the convex combination $\pi_3 = \frac{1}{2}(\pi_1 + \pi_2)$ is also optimal by linearity, and thus is supported on the graph of a function. Consequently, $T_1 = T_2$ and $\pi_1 = \pi_2$.
At last, note that any transport map between μ and ν of the type $T = \nabla\varphi$ is optimal, since its graph is c-cyclically monotone. □

Theorem 1.7.15. *(Brenier) The Kantorovich problem with quadratic cost $c(x, y) = \frac{1}{2}|x - y|^2$ on $\mathbb{R}^d$, and marginals $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ s.t. $\mu \ll \lambda$ admits a unique optimal coupling $\pi^* \in Cpl(\mu, \nu)$ of the form $\pi^* = (Id, \nabla\varphi)\mu$ for some convex function $\varphi : \mathbb{R}^d \to \mathbb{R}$. In particular, there exists a unique optimal transport map $T : \mathbb{R}^n \to \mathbb{R}^n$ from μ to ν of the form $T = \nabla\varphi$ for some convex function $\varphi : \mathbb{R}^n \to \mathbb{R}$.*

Proof. The thesis follows by observing that

$$\begin{aligned}
\inf_{\pi \in Cpl(\mu,\nu)} \int \frac{1}{2}|x - y|^2 \, d\pi &= \inf_{\pi \in Cpl(\mu,\nu)} \int \frac{1}{2}|x|^2 - x \cdot y + \frac{1}{2}|y|^2 \, d\pi \\
&= \inf_{\pi \in Cpl(\mu,\nu)} \left(\int |x|^2 \, d\mu + \int |y|^2 \, d\nu \right) + \int -x \cdot y \, d\pi \\
&= \frac{1}{2} \left(\int |x|^2 \, d\mu + \int |y|^2 \, d\nu \right) + \inf_{\pi \in Cpl(\mu,\nu)} \int -x \cdot y \, d\pi.
\end{aligned}$$

Then, the minimization problem for the quadratic cost $c(x, y) = \frac{1}{2}|x - y|^2$ is reduced to the minimization problem for the quadratic cost $c(x, y) = -x \cdot y$. □

1.8 Wasserstein Space

In this section, we introduce a metric on a subspace of the space $\mathcal{P}(X)$ of probability measures on a Polish space X, namely the space of p-finite moment probability measures.

Definition 1.8.1. A probability measure μ of a Polish space X has p-finite moment, with $p \geq 1$, if

$$\int_X d(x, x_0)^p \, d\mu(x) < \infty$$

for some $x_0 \in X$.

We will denote the space of finite p-moment probability measures on X by $\mathcal{P}_p(X)$.

Remark 1.8.2. The notion is well defined, in the sense that is does not depend on the choice of the point $x_0 \in X$. Indeed, for all $x_0, x_1 \in X$ we have

$$d(x, x_1)^p \leq (d(x, x_0) + d(x_0, x_1))^p \leq 2^{p-1}(d(x, x_0)^p + d(x_0, x_1)^p)$$

since $(a+b)^p \leq 2^{p-1}(a^p + b^p)$ for all $a, b \geq 0,\ p \geq 1$ by Jensen inequality with $t \mapsto t^p$. Therefore, if

$$\int_X d(x, x_0)^p \, d\mu(x) < \infty$$

for some $x_0 \in X$, and if $x_1 \in X$ is a distinct point, we get

$$\int_X d(x, x_1)^p \, d\mu(x) \leq 2^{p-1} \left(\int_X (d(x, x_0)^p \, d\mu(x) + d(x_0, x_1)^p \right) < \infty$$

Definition 1.8.3. The p-Wasserstein distance, with $p \geq 1$, between $\mu \in \mathcal{P}_p(X)$ and $\nu \in \mathcal{P}_p(Y)$ is given by

$$\mathcal{W}_p(\mu, \nu) := \left(\inf_{\pi \in Cpl(\mu, \nu)} \int_{X \times Y} d(x, y)^p \, d\pi(x, y) \right)^{1/p} = \left(\inf_{X \sim \mu, Y \sim \nu} \mathbb{E}[d(X, Y)^p] \right)^{1/p}$$

Remark 1.8.4. The p-Wasserstein distance between μ and ν is a Kantorovich optimal transport problem with $X = Y$ and metric cost $d(x, y)^p$. Hence, we have

$$\mathcal{W}_p(\mu, \nu) = P^K_{d^p}(\mu, \nu)^{1/p}.$$

In particular, the optimization problem defined by $\mathcal{W}_p(\mu, \nu)$ admits minimizers by continuity of the metric cost $c(x, y) = d(x, y)$.

We will prove in the following that $\mathcal{W}_p$ is well defined on $\mathcal{P}_p(X)$ in the sense that is assumes only finite values, and moreover we will prove that $\mathcal{W}_p$ is a metric, for any $p \geq 1$.

Lemma 1.8.5. *(Gluing) Let $\mu_1 \in \mathcal{P}(X_1),\ \mu_2 \in \mathcal{P}(X_2),\ \mu_3 \in \mathcal{P}(X_3)$ be probabilities on the Polish spaces X_1, X_2, X_3, and consider the couplings $\pi_{12} \in Cpl(\mu_1, \mu_2),\ \pi_{23} \in Cpl(\mu_2, \mu_3)$. Then, there exists a probability distribution $\pi \in \mathcal{P}(X_1 \times X_2 \times X_3)$ such that $p_{12}\pi = \pi_{12}$ and $p_{23}\pi = \pi_{23}$, where $p_{ij} : X_1 \times X_2 \times X_3 \to X_i \times X_j$ projection.*

Proof. Note that the couplings $\pi_{12} \in Cpl(\mu_1, \mu_2),\ \pi_{23} \in Cpl(\mu_2, \mu_3)$ can be disintegrated along their common marginal μ_2 as

$$\pi_{12}(dx_1, dx_2) = \pi_{12}(dx_1 \mid x_2) \; \mu_2(dx_2)$$
$$\pi_{12}(dx_2, dx_3) = \pi_{23}(dx_3 \mid x_2) \; \mu_2(dx_2)$$

Then, we can construct the 3-marginal probability measure π as

$$\pi(dx_1, dx_2, dx_3) := \pi_{12}(dx_1 \mid x_2)\ \mu_2(dx_2)\ \pi_{23}(dx_3 \mid x_2).$$

Hence, $\pi \in \mathcal{P}(X_1 \times X_2 \times X_3)$ has projections π_{12}, π_{23} by construction. $\square$

Theorem 1.8.6. *$(\mathcal{P}_p(X),\ \mathcal{W}_p)$ metric space, called p-Wasserstein space, for $p \geq 1$.*

Proof.
Well-def.) Let $\mu, \nu \in \mathcal{P}_p(X)$, let $\pi \in Cpl(\mu, \nu)$, and fix a point $x_0 \in X$. Thus, we get

$$\begin{aligned}
\mathcal{W}_p(\mu, \nu)^p &\leq \int d(x, y)^p\ d\pi(x, y) \\
&\leq 2^{p-1} \int d(x, x_0)^p\ d\pi(x, y) + 2^{p-1} \int d(x_0, y)^p\ d\pi(x, y) \\
&= 2^{p-1} \int d(x, x_0)^p\ d\mu(x) + 2^{p-1} \int d(x_0, y)^p\ d\nu(y) < +\infty,
\end{aligned}$$

since μ, ν have finite p-moment.
Symmetry) Let $\pi \in Cpl(\mu, \nu)$, and consider the transposed coupling $\pi^T := T\pi$ obtained as pushforward of π under the map $T(x, y) = T(y, x)$. Note, that $\pi^T \in Cpl(\nu, \mu)$ since

$$p_1\pi^T = p_1 T\pi = p_2\pi = \nu, \quad p_2\pi^T = p_2 T\pi = p_1\pi = \mu.$$

Thus, we have

$$\begin{aligned}
\int d(x, y)^p\ d\pi(x, y) &= \int d(T(x, y))^p\ d\pi(x, y) \quad \text{by symmetry of } d \\
&= \int d(x, y)^p\ d\pi^T(x, y) \quad \text{by change of variable, since } T\pi = \pi^T.
\end{aligned}$$

Therefore every coupling of (μ, ν) gives a coupling of (ν, μ) with exactly the same cost, and by taking infimum over all couplings, we get

$$\inf_{\pi \in Cpl(\mu,\nu)} \int_{X \times Y} d(x, y)^p\ d\pi(x, y) \leq \inf_{\pi \in Cpl(\nu,\mu)} \int_{X \times Y} d(x, y)^p\ d\pi(x, y).$$

The symmetry follows by changing the roles of μ, ν in the inequality.
Non-Negativity) Let $\mu \in \mathcal{P}_p(X)$ and note that the diagonal coupling $\pi := (Id, Id)\pi$ has cost

$$\int d(x, y)^p\ d\pi(x, y) = \int d(x, x)^p\ d\mu(x) = 0 \quad \text{by change of variable}$$

since $d = 0$ on the diagonal of $X \times X$. Then, we have

$$0 \leq \mathcal{W}_p(\mu, \mu)^p = \inf_{\pi \in Cpl(\mu,\mu)} \int d(x, y)^p\ d\pi(x, y) \leq 0.$$

Conversely, suppose that

$$\mathcal{W}_p(\mu, \nu)^p = \inf_{\pi \in Cpl(\mu,\nu)} \int d(x, y)^p\ d\pi(x, y) = 0$$

for some $\mu, \nu \in \mathcal{P}_p(X)$. Hence, $d(x, y) = 0$ π-a.s. for some $\pi \in Cpl(\mu, \nu)$, which implies that $x = y$ π-a.e., since d non-negative. Therefore, π is entirely concentrated on the

diagonal of $X \times X$, i.e. π induced by identity map and $\nu = (Id, Id)\mu = \mu$.
Triangle inequality) Let $\mu_1, \mu_2, \mu_3 \in \mathcal{P}_p(X)$, and let $\pi_{12} \in Cpl(\mu_1, \mu_2)$, $\pi_{23} \in Cpl(\mu_2, \mu_3)$ be optimal coupling for the respective Wasserstein problems.
Then, there exists a probability distribution $\pi \in \mathcal{P}(X_1 \times X_2 \times X_3)$ such that $p_{12}\pi = \pi_{12}$ and $p_{23}\pi = \pi_{23}$, by gluing lemma. Now, let (Z_1, Z_2, Z_3) be a random vector with law π, such that $(Z_1, Z_2) \sim \pi_{12}$ and $(Z_2, Z_3) \sim \pi_{23}$ by construction. Finally, we get

$$\begin{aligned}
\mathcal{W}_p(\mu_1, \mu_3) &= \left(\inf_{X \sim \mu_1, Y \sim \nu_3} \mathbb{E}[d(X, Y)^p] \right)^{1/p} \\
&\leq \mathbb{E}[d(Z_1, Z_3)^p]^{1/p} \quad \text{since } Z_1 \sim \mu_1,\ Z_3 \sim \mu_3 \\
&\leq \mathbb{E}[d(Z_1, Z_2)^p]^{1/p} + \mathbb{E}[d(Z_2, Z_3)^p]^{1/p} \quad \text{by Minkowski inequality} \\
&= \mathcal{W}_p(\mu_1, \mu_2) + \mathcal{W}_p(\mu_2, \mu_3),
\end{aligned}$$

since $\pi_{12} \sim (Z_1, Z_2)$, $\pi_{23} \sim (Z_2, Z_3)$ optimal couplings. □

Remark 1.8.7. For $p = 1$, we have by Kantorovich-Rubenstein formula

$$\mathcal{W}_1(\mu, \nu) = \inf_{\pi \in Cpl(\mu,\nu)} \int d(x, y)\ d\pi(x, y) = \sup_{\varphi \text{ 1-Lip}} \int \varphi\ d(\mu - \nu).$$

The Wasserstein distance $\mathcal{W}_1$ of order 1 is sometimes called the Kantorovich-Rubenstein distance on $\mathcal{P}_1(X)$.

Example 1.8.8. The p-Wasserstein distance between Dirac measures coincides with the usual metric for any $p \geq 1$. Indeed, the space of couplings $Cpl(\delta_x, \delta_y)$ consists only of the independent coupling $\delta_{(x,y)} = \delta_x \otimes \delta_y$ by Example 1.2.5, and we get

$$\mathcal{W}_p(\delta_x, \delta_y) = \left(\int d(x, y)\ d(\delta_x \otimes \delta_y)(x, y) \right)^{1/p} = d(x, y).$$

Proposition 1.8.9. *$\mathcal{W}_p \leq \mathcal{W}_q$ for $1 \leq p \leq q < +\infty$.*

Proof. Let $r := q/p$ and $r' := q/(q - p)$ such that $r, r' \geq 1$, $1/r + 1/r' = 1$. Then, by Holder inequality we get

$$||d(X, Y)^p \cdot 1||_1 \leq ||d(X, Y)^p||_r \cdot ||1||_{r'} = \mathbb{E}[d(X, Y)^{rp}]^{1/r} = \mathbb{E}[d(X, Y)^q]^{p/q}.$$

Then, we have

$$\mathcal{W}_p(\mu, \nu)^p = \inf_{X \sim \mu,\ Y \sim \nu} \mathbb{E}[d(X, Y)^p] \leq \inf_{X \sim \mu, Y \sim \nu} \mathbb{E}[d(X, Y)^q]^{p/q} = \mathcal{W}_q(\mu, \nu)^p.$$

□

Remark 1.8.10. Up to now, we have introduced two a priori distinct topologies on the space $\mathcal{P}_p(X)$:

- The weak topology induced by weak convergence on the subspace $\mathcal{P}_p(X) \subseteq \mathcal{P}(X)$
- The metric topology induced by the p-Wasserstein distance

We will now investigate the relation between these two notions of convergence.

Theorem 1.8.11. *(Skorohod representation) Let $\{\mu_n\} \subseteq \mathcal{P}(X)$ be a sequence of probability measures on a Polish space X converging weakly to a probability $\mu \in \mathcal{P}(X)$. Then, there exist a sequence $\{Z_n\}_n$ of X-valued random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that*

(i) $X_n \to X$ almost surely for $n \to \infty$.

(ii) $X_n \sim \mu_n, \ X \sim \mu$ for all $n \in \mathbb{N}$.

Theorem 1.8.12. *Let X be a bounded Polish space. Then, $\mathcal{W}_p$ metrizes the weak topology of $\mathcal{P}(X)$, for $p \geq 1$, i.e.*

$$\mu_n \to \mu \text{ weakly in } \mathcal{P}(X) \iff \mathcal{W}_p(\mu_n, \mu) \to 0.$$

Proof. Let $\{\mu_n\} \subseteq \mathcal{P}(X)$ be a sequence converging weakly to $\mu \in \mathcal{P}(X)$. Then, there exist $\{Z_n\}_n, Z$ random variables with laws μ_n, μ respectively, such that

$$Z_n \to Z \quad \mathbb{P}\text{-almost surely for } n \to \infty.$$

by Skorohod representation theorem. Let $\pi_n \in Cpl(\mu_n, \mu)$ be the law of the vector (Z_n, Z), and note that

$$\begin{aligned}
\limsup_{n\to\infty} \mathcal{W}_p(\mu_n, \mu)^p &= \limsup_{n\to\infty} \inf_{\pi \in Cpl(\mu_n, \nu)} \int d(x,y) \ d\pi(x,y) \\
&\leq \limsup_{n\to\infty} \int d(x,y) \ d\pi_n(x,y) \\
&= \limsup_{n\to\infty} \mathbb{E}[d(Z_n, Z)] \quad \text{since } \pi_n \sim (Z_n, Z) \\
&= 0 \quad \text{by DCT.}
\end{aligned}$$

Conversely, suppose that $\mathcal{W}_p(\mu_n, \mu) \to 0$ for $n \to \infty$, and let $f : X \to \mathbb{R}$ be a 1-Lipschitz map. Then, we have

$$\begin{aligned}
\left| \int f \ d\mu_n - \int f \ d\mu \right| &\leq \sup_{f \text{ 1-Lip}} \int f \ d(\mu_n - \mu) \\
&= \inf_{\pi \in Cpl(\mu_n, \mu)} \int d \ d\pi \quad \text{by Kantorovich-Rubenstein} \\
&= \mathcal{W}_1(\mu_n, \mu) \leq \mathcal{W}_p(\mu_n, \mu) \to 0
\end{aligned}$$

by Proposition 1.8.9. The thesis follows by observing that on bounded Polish spaces X, the weak convergence in $\mathcal{P}(X)$ by continuous, bounded test functions is equivalent to the weak convergence by 1-Lipschitz test functions. $\square$

We will now remove the boundedness hypothesis on the previous theorem, and characterize completely the topology metrized by the Wasserstein distance.

Definition 1.8.13. A sequence of measures $\{\mu_n\}_n \subseteq \mathcal{P}_p(X)$ converges weakly to a measure $\mu \in \mathcal{P}_p(X)$ if $\mu_n \to \mu$ weakly in $\mathcal{P}(X)$ and

$$\int d(x, x_0)^p \ d\mu_n \underset{n\to\infty}{\longrightarrow} \int d(x, x_0)^p \ d\mu.$$

for some $x_0 \in X$.

There exists many equivalent ways of introducing a notion of convergence in $\mathcal{P}_p(X)$.

Proposition 1.8.14. *Let $\{\mu_n\}_n \subseteq \mathcal{P}_p(X)$ and $\mu \in \mathcal{P}_p(X)$. Then, the following are equivalent:*

(i) $\mu_n \to \mu$ *weakly in* $\mathcal{P}_p(X)$

(ii) $\mu_n \to \mu$ *weakly in* $\mathcal{P}(X)$*, and*

$$\limsup_{n\to\infty} \int d(x,x)^p \; d\mu_n(x) \leq \int d(x,x_0)^p \; d\mu.$$

(iii) $\mu_n \to \mu$ *weakly in* $\mathcal{P}(X)$ *and*

$$\lim_{R\to\infty} \limsup_{n\to\infty} \int_{d(x,x_0)\geq R} d(x,x_0)^p \; d\mu(x) = 0.$$

(iv) If $f : X \to \mathbb{R}$ *continuous such that* $|f(x)| \leq K(1 + d(x,x_0)^p)$ *for some* $K \in \mathbb{R}$*, then*

$$\int f \; d\mu_n \to \int f \; d\mu.$$

Lemma 1.8.15. *Let $\{\mu_n\}_n \subseteq \mathcal{P}_p(X)$ be a Cauchy sequence respect to $\mathcal{W}_p$.*
Then, $\{\mu_n\}_n$ is tight

Proof. Lemma 6.14, [7]. □

Theorem 1.8.16. *$\mathcal{W}_p$ metrizes the weak convergence in $\mathcal{P}_p(X)$, for $p \geq 1$, i.e.*

$$\mu_n \to \mu \text{ weakly in } \mathcal{P}_p(X) \iff \mathcal{W}_p(\mu_n, \mu) \to 0.$$

Proof. Theorem 6.9, [7]. □

Theorem 1.8.17. *Let X be a Polish space, then $(\mathcal{P}_p(X), \mathcal{W}_p)$ Polish space for $p \geq 1$.*

Proof. Theorem 6.18, [7]. □

1.9 Displacement Interpolation

In this section, we discuss the idea that leads to the dynamic formulation of optimal transport, namely the displacement interpolation.
Recall that, in $X = Y = \mathbb{R}^d$, there exists a unique constant speed geodesic between any couple of fixed points $x_0, x_1 \in \mathbb{R}^d$ given by

$$x_t := (1-t)x_0 + tx_1, \quad \text{for } t \in [0,1].$$

Explicitly, x_t is the unique optimizer of the problem

$$\inf_{\gamma_0 = x_0, \gamma_1 = x_1} \int_0^1 |\dot{\gamma}_t|^2 \; dt.$$

In the following, we will generalize this result to constant speed geodesics in $\mathcal{P}_p(\mathbb{R}^d)$.

Proposition 1.9.1. *Let $\mu_0, \mu_1 \in \mathcal{P}_p(\mathbb{R}^d)$, and X_0, X_1 $\mathbb{R}^d$-valued variables with laws μ_0, μ_1. Then, the curve $\{\mu_t\}_{t\in[0,1]} \subseteq \mathcal{P}_p(\mathbb{R}^d)$ given by*

$$\mu_t \sim (1-t)X_0 + tX_1$$

is a constant speed geodesic in $\mathcal{P}_p(\mathbb{R}^d)$ if and only if (X_0, X_1) is $\mathcal{W}_p$-optimal.

Proof. Let $\mu_0, \mu_1 \in \mathcal{P}_p(\mathbb{R}^d)$, and let $\pi \in Cpl(\mu_0, \mu_1)$ be a $\mathcal{W}_p$-optimal coupling, i.e.

$$\mathcal{W}_p(\mu_0, \mu_1)^p = \int_{\mathbb{R}^d \times \mathbb{R}^d} |x-y|^p \, d\pi(x,y).$$

Now, consider X_0, X_1 $\mathbb{R}^d$-valued random variables such that $(X_0, X_1) \sim \pi$ and, define for all $t \in [0,1]$

$$X_t := (1-t)X_0 + tX_1,$$

with law μ_t. Then, we have by construction

$$\begin{aligned} \mathcal{W}_p(\mu_s, \mu_t) &= \left(\inf_{X\sim\mu_s, Y\sim\mu_t} \mathbb{E}[|X-Y|^p] \right)^{1/p} \\ &\leq \mathbb{E}[|X_s - X_t|^p]^{1/p} \\ &= \mathbb{E}[|s-t|^p \; |X_0 - X_1|^p]^{1/p} \\ &= |t-s| \; \mathcal{W}_p(\mu_0, \mu_1), \end{aligned}$$

since (X_0, X_1) optimal coupling for $\mathcal{W}_p(\mu_0, \mu_1)$. Then, the curve $\{\mu_t\}_{t\in[0,1]}$ is a constant speed geodesic between μ_0 and μ_1.
Conversely, all the constant speed geodesics between μ_0, μ_1 are of this type. □

Remark 1.9.2. In particular, the uniqueness of geodesic is not guaranteed in general, as the Wasserstein problem may might admit non-unique optimizers.

Example 1.9.3. The unique constant speed geodesics between $\delta_x, \delta_y \in \mathcal{P}_p(\mathbb{R}^d)$ is the curve $\{\delta_{x_t}\}_{t\in[0,1]}$ with

$$x_t := (1-t)x + ty.$$

Indeed, we have

$$\mathcal{W}_p(\delta_{x_s}, \delta_{x_t}) = |x_s - x_t| = |t-s| \; |x-y|$$

since the Wasserstein distance coincides with the underlying metric for Dirac measures. However, note that the curve obtained by convex interpolation $\mu_t := (1-t)\delta_x + t\delta_y$ is not a constant speed geodesics from δ_x to δ_y. Hence, it holds

$$\mathcal{W}_p(\mu_s, \mu_t) = |s-t|^{1/p} \; |x-y| > |t-s||x-y| = |t-s| \; \mathcal{W}_p(\delta_x, \delta_y).$$

Intuitively, the constant speed geodesic δ_{x_t} moves from δ_x to δ_y by changing its support, while the convex interpolation μ_t goes from δ_x to δ_y by moving mass from x to y.

As mentioned before, in the deterministic setting, the curve $x_t = (1-t)x_0 + tx_1$ is the unique optimizer of the problem

$$\inf_{\gamma_0 = x_0, \gamma_1 = x_1} \int_0^1 |\dot{\gamma}_t|^2 \, dt.$$

A similar result is guaranteed by the Brenier's theorem in the case of constant speed geodesics in $\mathcal{P}_2(\mathbb{R}^d)$.

Proposition 1.9.4. *The constant speed and absolutely continuous geodesics between $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, with $\mu \ll \lambda$, are the laws of the optimizers of the problem*

$$\mathcal{W}_2(\mu,\nu)^2 = \inf_{\substack{\{X_t\}_t \in AC(\mathbb{R}^d) \\ X_0\sim\mu,\ X_1\sim\nu}} \int_0^1 \mathbb{E}[\dot{X}_t^2]\ dt,$$

where $AC(\mathbb{R}^d)$ is the space of process with absolutely continuous paths $t \mapsto X_t \in \mathbb{R}^d$.

Proof. Let $\{\mu_t\}_t$ be a constant speed geodesic between μ, ν, i.e.

$$\mu_t \sim X_t := (1-t)X_0 + tX_1 \quad \text{for } t \in [0,1],$$

where (X_0, X_1) is $\mathcal{W}_2(\mu,\nu)$-optimal, and $t \mapsto X_t$ is absolutely continuous.
Now, note that the 2-Wasserstein problem coincides with the Kantorovich problem with quadratic cost $c(x,y) = \frac{1}{2}|x-y|^2$, since

$$\mathcal{W}_2(\mu,\nu)^2 = \inf_{\pi\in Cpl(\mu,\nu)} \int_{\mathbb{R}^d\times\mathbb{R}^d} |x-y|^2\ d\pi(x,y) = \inf_{\pi\in Cpl(\mu,\nu)} \int_{\mathbb{R}^d\times\mathbb{R}^d} \frac{1}{2}|x-y|^2\ d\pi(x,y).$$

Then, there exists a unique optimal transport map from μ to ν of the form $\nabla\varphi : \mathbb{R}^d \to \mathbb{R}^d$, for some convex function $\varphi : \mathbb{R}^d \to \mathbb{R}$, by Brenier's theorem, i.e. it holds

$$\nabla\varphi(\mu) = \nu.$$

Equivalently, since $X_0 \sim \mu$ and $X_1 \sim \nu$, we also have $\nabla\varphi(X_0) = X_1$, which implies $X_t = (1-t)X_0 + t\nabla\varphi(X_0)$ with

$$\dot{X}_t = \nabla\varphi(X_0) - X_0.$$

Finally, we get

$$\mathcal{W}_2(\mu,\nu)^2 = \mathbb{E}[|X_0 - X_1|^2] = \int_0^1 \mathbb{E}[\dot{X}_t^2]\ dt.$$

Moreover, if $\{X_t\}_{t\in[0,1]}$ is an absolutely continuous curve with $X_0 \sim \mu$ and $X_1 \sim \nu$, then we have

$$\begin{aligned}
\mathcal{W}_2(\mu,\nu)^2 &= \inf_{X\sim\mu,\ Y\sim\nu} \mathbb{E}[|X-Y|^2] \le \mathbb{E}[|X_0 - X_1|^2] \\
&\le \mathbb{E}\left[\left|\int_0^1 \dot{X}_t\ dt\right|^2\right] \le \mathbb{E}\left[\int_0^1 \dot{X}_t^2\ dt\right] \quad \text{by Jensen} \\
&= \int_0^1 \mathbb{E}[\dot{X}_t^2]\ dt \quad \text{by Fubini.}
\end{aligned}$$

The thesis follows by taking the infimum over all absolutely continuous curves with $X_0 \sim \mu,\ X_1 \sim \nu$. □

1.10 Multi-marginal Optimal Transport

In this section, we briefly discuss a multidimensional generalization of the optimal transport problem in the Kantorovich formulation, originally introduced in [3]. We then present the Wasserstein barycenter problem as a particular instance of a multi-marginal optimal transport problem.

Definition 1.10.1. A joint probability measure of $\mu_1, \ldots, \mu_n \in \mathcal{P}(X)$ is a probability measure $\pi \in \mathcal{P}(X^n)$ with marginals $\mu_1, \ldots, \mu_n$, i.e.

$$p_i \pi = \mu_i \text{ for } i = 1, \ldots, n$$

where $p_i : X^i \to X$ is the projection on the i-th component.

We will denote the space of joint probabilities of $\mu_1, \ldots, \mu_n$ by $\Pi(\mu_1, \ldots, \mu_n)$.

Definition 1.10.2. A cost function for an optimal transport problem with marginals $\mu_1, \ldots, \mu_n \in \mathcal{P}(X)$ is a measurable map $c : X^n \to \mathbb{R} \cup \{+\infty\}$ such that

$$c(x_1, \ldots, x_n) \geq \sum_{i=0}^{n} a_i(x_i) \text{ for all } x_1, \ldots, x_n \in X$$

for some functions $a_1, \ldots, a_n : X \to \mathbb{R} \cup \{-\infty\}$ with $a_i \in L^1(\mu_i)$ for $i = 1, \ldots, n$.

Again, the lower bound assumption $c(x_1, \ldots, x_n) \geq \sum_{i=0}^{n} a_i(x_i)$ guarantees that the expected cost $\mathbb{E}[c(X_1, \ldots, X_n)]$ is well defined in $\mathbb{R} \cup \{+\infty\}$, with $X_i \sim \mu_i$.

Definition 1.10.3. The n-marginal Kantorovich functional with cost $c : X^n \to \mathbb{R} \cup \{+\infty\}$ is the map $V_c : \mathcal{P}(X)^n \to \mathbb{R} \cup \{+\infty\}$ such that

$$V_c^n(\mu_1, \ldots, \mu_n) := \inf_{\pi \in \Pi(\mu_1, \ldots, \mu_n)} \int_{X^n} c \, d\pi.$$

Definition 1.10.4. The n-marginal dual Kantorovich functional with cost $c : X^n \to \mathbb{R} \cup \{+\infty\}$ is the map $D_c^n(\mu_1, \ldots, \mu_n) : \mathcal{P}(X)^n \to \mathbb{R} \cup \{+\infty\}$ such that

$$D_c^n(\mu_1, \ldots, \mu_n) := \sup_{(\varphi_1, \ldots, \varphi_n) \in D^n(c)} \sum_{i=1}^{n} \int_X \varphi_i \, d\mu_i$$

with $D^n(c) := \{(\varphi_1, \ldots, \varphi_n) \in L^1(\mu_1) \times \cdots \times L^1(\mu_n) \mid c \geq \bigoplus_{i=1}^{n} \varphi_i\}$ research space.

The classical existence and duality results for 2-marginal optimal transport (Theorem 1.5.4, Theorem 1.6.5) can be adapted to the n-dimensional setting.

Theorem 1.10.5. *Let $c : X^n \to \mathbb{R}$ be a LSC cost function for a n-dimensional optimal transport problem with marginals $\mu_1, \ldots, \mu_n \in \mathcal{P}(X)$. Then, the following holds:*

(i) If $c \geq 0$, then the map

$$\pi \mapsto \int_{X^n} c \, d\pi$$

is lower-semicontinuous.

(ii) There exists a minimizer for the n-marginal Kantorovich problem, i.e.

$$\exists \ \pi^* \in argmin_{\pi \in \Pi(\mu_1, \ldots, \mu_n)} \int_{X^n} c \, d\pi.$$

(iii) Duality holds, i.e. $D_c^n(\mu_1, \ldots, \mu_n) = V_c^n(\mu_1, \ldots, \mu_n)$.

Remark 1.10.6. The geometry of optimal joint measures becomes substantially richer for $n \geq 3$, and understanding their structure remains an active area of research. Indeed, uniqueness of optimizers and concentration on the graph of a transport map are no longer typical properties of multi-dimensional optimal transport problems.

A particular instance of a multi-marginal Kantorovich problem is given by the Wasserstein barycenter problem, firstly introduced in 2011 by [1].

Definition 1.10.7. The p-Wasserstein barycenter functional is the map $V^W : \mathcal{P}_p(\mathbb{R}^d)^N \to \mathbb{R} \cup \{+\infty\}$ such that

$$V^W(\mu_1, \dots, \mu_n) := \inf_{\mu \in \mathcal{P}_p(\mathbb{R}^d)} \frac{1}{N} \sum_{i=1}^{N} \mathcal{W}_p(\mu_i, \mu)^p.$$

In particular, a Wasserstein barycenter of $\mu_1, \dots, \mu_n \in \mathcal{P}_p(\mathbb{R}^d)$ is a minimizer of the optimization problem.

Remark 1.10.8. Intuitively, a Wasserstein barycenter of $\mu_1, \dots, \mu_N \in \mathcal{P}_p(X)$ is a notion of mean in the Wasserstein space $(\mathcal{P}_p(\mathbb{R}^d), \mathcal{W}_p)$. Moreover, in general the Wasserstein barycenter is not unique, reflecting the geometric complexity of $\mathcal{P}_p(X)$.

The next result shows that we can regard the 2-Wasserstein barycenter problem as a multi-marginal Kantorovich problem for a specific quadratic cost function.

Theorem 1.10.9. *Let $\mu_1, \dots, \mu_n \in \mathcal{P}_2(\mathbb{R}^d)$ be probability measures with finite second moment. Then, the 2-Wasserstein barycenter problem is a multi-marginal Kantorovich problem with cost $c : (\mathbb{R}^d)^N \to \mathbb{R}$ such that $c(x_1, \dots, x_N) := \frac{1}{N}\sum_{i=1}^N \|x_i - T(x_1, \dots, x_n)\|^2$, i.e.*

$$V^W(\mu_1, \dots, \mu_n) := \inf_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} \frac{1}{N} \sum_{i=1}^{N} \mathcal{W}_2(\mu_i, \mu)^2 = V_c^N(\mu_1, \dots, \mu_N).$$

where $T(x_1, \dots, \mu_N) = \bar{x}_N = \frac{1}{N}\sum_{i=1}^N x_i \in \mathbb{R}^d$ average point.
Moreover, if π^* is an optimizer of $V_c^N(\mu_1, \dots, \mu_N)$, then $T\pi^*$ is a 2-Wasserstein barycenter of $\mu_1, \dots, \mu_N$.

Proof. We have

$$\begin{aligned}\inf_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} \frac{1}{N} \sum_{i=1}^{N} \mathcal{W}_2(\mu_i, \mu)^2 &= \inf_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} \frac{1}{N} \sum_{i=1}^{N} \inf_{X_i \sim \mu_i, X \sim \mu} \mathbb{E}[\|X_i - X\|^2] = \\ &= \inf_{X_1 \sim \mu_1, \dots, X_n \sim \mu_N} \inf_{\mu \in \mathcal{P}_2(\mathbb{R}^d), X \ \sim \mu} \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}[\|X_i - X\|^2] = \\ &= \inf_{X_1 \sim \mu_1, \dots, X_n \sim \mu_N} \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}[\|X_i - \bar{X}_N\|^2]\end{aligned}$$

since the sample mean of the data points minimizes the squared loss. Then, we obtain

$$\begin{aligned}
\inf_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} \frac{1}{N} \sum_{i=1}^{N} \mathcal{W}_2(\mu_i, \mu)^2 &= \inf_{X_1 \sim \mu_1, \ldots, X_n \sim \mu_N} \mathbb{E}\left[\frac{1}{N} \sum_{i=1}^{N} \|X_i - \bar{X}_N\|^2 \right] \\
&= \inf_{\pi \in \Pi(\mu_1, \ldots, \mu_n)} \int_{(\mathbb{R}^d)^N} \frac{1}{N} \sum_{i=1}^{N} \|x_i - \bar{x}_N\|^2 \; d\pi(x_1, \ldots, x_N) \\
&= \inf_{\pi \in \Pi(\mu_1, \ldots, \mu_n)} \int_{(\mathbb{R}^d)^N} c(x_1, \ldots, x_N) \; d\pi(x_1, \ldots, x_N) \\
&= V_c^n(\mu_1, \ldots, \mu_N).
\end{aligned}$$

Thus, $W(\mu_1, \ldots, \mu_N)$ is a multi-marginal Kantorovich problem with quadratic cost c. Moreover, by Theorem 1.10.5 the infimum is attained by some optimizer π^* and the induced measure $T\pi^*$ on $\mathbb{R}^d$ by pushforward under the map $T : (\mathbb{R}^d)^N \to \mathbb{R}^d$, is a 2-Wasserstein barycenter of $\mu_1, \ldots, \mu_N$. $\square$

2 Weak Optimal Transport (WOT)

2.1 Introduction

In this section, we generalize the notion of optimal transport problem by considering a broader class of admissible costs. Firstly, we recall some results about the disintegration of measures.

Definition 2.1.1. A transition kernel from a measurable space $(E, \mathcal{E})$ to a measurable space $(F, \mathcal{F})$ is a map $K : E \times \mathcal{F} \to [0, 1]$ such that:

1. $x \mapsto K(x, A)$ is $\mathcal{E}$-measurable for all $A \in \mathcal{F}$.
2. $B \mapsto K(x, B)$ is a probability measure on $(F, \mathcal{F})$ for all $x \in E$.

Theorem 2.1.2. *Let μ be a probability measure on $(E, \mathcal{E})$, and K a transition kernel from $(E, \mathcal{E})$ to $(F, \mathcal{F})$. Then, there exists a unique probability measure $\mu \otimes K$ on $(E \times F, \mathcal{E} \otimes \mathcal{F})$ such that*

$$(\mu \otimes K)(A \times B) = \int_A K(x, B) d\mu(x) \quad \forall A \in \mathcal{E},\ B \in \mathcal{F}.$$

Theorem 2.1.3. *(Disintegration) Let X, Y be Polish spaces, and π a probability measure on $X \times Y$ with first marginal $\mu \in \mathcal{P}(X)$. Then, there exists a family $\{\pi_x\}_{x \in X}$ of probability measures on Y, called disintegration of π respect to μ, such that:*

1. *$x \mapsto \int_Y f(x, y) d\pi_x(y)$ Borel measurable for μ-a.s. $x \in X$.*
2. *For any $f : X \times Y \to \mathbb{R}$ bounded, Borel measurable, it holds*

$$\int_{X \times Y} f(x, y) d\pi(x, y) = \int_X \left(\int_Y f(x, y) d\pi_x(y) \right) d\mu(x)$$

Moreover, the family $\{\pi_x\}_{x \in X}$ is essentially unique, i.e. for any other family $\{\tilde{\pi}_x\}_{x \in X}$ it holds $\pi_x = \tilde{\pi}_x$ for μ-a.e. $x \in X$.

The weak optimal transport problem was first introduced by Gozlan, Roberto, Samson and Tetali [4], as consequence of the following observation.
Consider the Kantorovich optimal transport problem with marginals $\mu \in \mathcal{P}(X),\ \nu \in \mathcal{P}(Y)$ and cost function $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$. Then, we have

$$\begin{aligned} P_c^K(\mu, \nu) &= \inf_{\pi \in Cpl(\mu, \nu)} \int_{X \times Y} c(x, y) d\pi(x, y) = \\ &= \inf_{\pi \in Cpl(\mu, \nu)} \int_X \int_Y c(x, y)\ d\pi_x(y)\ d\mu(x) \qquad \text{by disintegration of } \pi \text{ by } \mu \\ &= \inf_{\pi \in Cpl(\mu, \nu)} \int_X C(x, \pi_x)\ d\mu(x) \end{aligned}$$

where $C(x, \rho) := \int_Y c(x, y)\ d\rho(y)$ weak cost functional for $\rho \in \mathcal{P}(Y)$.
Thus, we may define a notion of weak optimal transport by enlarging the class of admissible cost functions $c(x, y)$ to costs of the type $C(x, \rho)$ for $x \in X$ and $\rho \in \mathcal{P}(Y)$.

Definition 2.1.4. A cost function for a weak optimal transport problem with distributions $\mu \in \mathcal{P}(X),\ \nu \in \mathcal{P}_p(Y)$ is a measurable map $C : X \times \mathcal{P}_p(Y) \to \mathbb{R} \cup \{+\infty\}$.

Definition 2.1.5. The weak optimal transport functional with cost $C : X \times \mathcal{P}_p(Y) \to \mathbb{R} \cup \{+\infty\}$ is the map $V_C^{WOT} : \mathcal{P}(X) \times \mathcal{P}_p(Y) \to \mathbb{R} \cup \{+\infty\}$ such that

$$V_C^{WOT}(\mu, \nu) := \inf_{\pi \in Cpl(\mu,\nu)} \int_X C(x, \pi_x)\ d\mu(x) \quad \text{(WKP)}$$

where $\{\pi_x\}_{x \in X} \subseteq \mathcal{P}(Y)$ is the family of probability measures on Y obtained by disintegration of π respect to μ.

Lemma 2.1.6. *Let $\pi \in \mathcal{P}(X \times Y)$ be a probability measure with marginals $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$ such that $\pi \ll \mu \otimes \nu$. Then, the following hold:*

(i) $\pi_x \ll \nu$ for μ-a.s. $x \in X$

(ii) The Radon-Nikodym of π_x respect to ν is given by

$$\frac{d\pi_x}{d\nu}(y) = \frac{d\pi}{d\mu \otimes \nu}(x, y) \quad \textit{for } \mu \otimes \nu\textit{-a.s } (x, y) \in X \times Y.$$

Example 2.1.7. *Entropic Optimal Transport*
In many application of classical optimal transport, some kind of regularization is required in order to implement efficient algorithms with provable convergence. The most popular choice in this context is entropic regularization, as it allows for Sinkhorn's algorithm, that can be implemented at large scale and is analytically tractable.
The entropic optimal transport functional with cost $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ is the map $V_c^{EOT} : \mathcal{P}(X) \times \mathcal{P}(Y) \to \mathbb{R} \cup \{+\infty\}$ such that

$$V_c^{EOT}(\mu, \nu) := \inf_{\pi \in Cpl(\mu,\nu)} \int_{X \times Y} c\ d\pi + \varepsilon H(\pi \mid \mu \otimes \nu),$$

where $\varepsilon > 0$, and $H : \mathcal{P}(X) \times \mathcal{P}(Y) \to \mathbb{R} \cup \{+\infty\}$ is the relative entropy given by

$$H(\nu \mid \rho) := \begin{cases} \int_X \log\left(\frac{d\nu}{d\rho}(x)\right) d\nu(x) & \eta \ll \nu, \\ +\infty & \text{otherwise.} \end{cases}$$

The basic idea is to solve the problem for arbitrary small $\varepsilon > 0$ to obtain an approximation of the unregularized optimal transport problem , corresponding to $\varepsilon = 0$, and then consider the optimizer for $\varepsilon \to 0$.
Moreover, note the entropic optimal transport problem can be interpreted as a weak optimal transport problem. Indeed, for $\pi \ll \mu \otimes \nu$ we get by disintegration theorem

$$\begin{aligned} H(\pi \mid \mu \otimes \nu) &= \int_{X \times Y} \log\left(\frac{d\pi}{d\mu \otimes \nu}(x, y)\right) d\pi(x, y) = \\ &= \int_X \int_Y \log\left(\frac{d\pi_x}{d\nu}(y)\right)\ d\pi_x(y)\ d\mu(x) = \\ &= \int_X H(\pi_x \mid \nu)\ d\mu(x) \end{aligned}$$

Then, we have

$$\begin{aligned} V_c^{EOT}(\mu, \nu) &= \inf_{\pi \in Cpl(\mu,\nu)} \int_{X \times Y} c\ d\pi + \varepsilon H(\pi \mid \mu \otimes \nu) = \\ &= \inf_{\pi \in Cpl(\mu,\nu)} \int_X \int_Y c(x, y)\ d\pi_x(y)\ d\mu(x) + \varepsilon \int_X H(\pi_x \mid \nu)\ d\mu(x) = \\ &= \inf_{\pi \in Cpl(\mu,\nu)} \int_X C(x, \pi_x)\ d\mu(x) = V_C^{WOT}(\mu, \nu), \end{aligned}$$

where $C(x,\rho) = \varepsilon H(\rho \mid \nu) + \int_Y c(x,y)\ d\rho(y)$ is the associated weak cost function. For a more detailed treatment of this topic, see [5].

2.2 Existence, duality and structural results

In this section, we want to adapt classical optimal transport results to the weak setting. In particular, we want to answer to the following questions:

- Does (WKP) admit an optimizer? Is it unique?
- Does duality hold for (WKP)?
- Is there any structure in the optimizers of (WKP)?

Recall that in the classical setting, the crucial point for the existence of optimizers to the Kantorovich problem was the lower-semicontinuity property of the map

$$\pi \mapsto \int_{X\times Y} c\ d\pi,$$

which was guaranteed by requiring the cost $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ to be bounded below and lower-semicontinuous.
However, in the weak setting the implication does not hold anymore, and the map

$$\pi \mapsto \int_X C(x,\pi_x)\ d\mu(x)$$

is in general not lower-semicontinuous, even if the cost C is continuous on $X \times \mathcal{P}_p(Y)$.

Theorem 2.2.1. *(Existence) Let $C : X \times \mathcal{P}_p(Y) \to \mathbb{R} \cup \{+\infty\}$ be a jointly lower-semicontinuous with respect to the product topology on $X \times \mathcal{P}_p(Y)$, bounded below and convex in the second argument, cost function for a weak optimal transport problem with marginals $\mu \in \mathcal{P}(X),\ \nu \in \mathcal{P}_p(Y)$. Then, the following hold:*

(i) The map $\pi \mapsto \int_X C(x,\pi_x)\ d\mu(x)$ is lower-semicontinuous.

(ii) There exists an optimizer to (WKP), i.e.

$$\exists\ \pi^* \in argmin_{\pi\in Cpl(\mu,\nu)} \int_X C(x,\pi_x)\ d\mu(x).$$

Proof. Theorem 1.1, [2]. □

Theorem 2.2.2. *(Duality) Let $C : X\times\mathcal{P}_p(Y) \to \mathbb{R}\cup\{+\infty\}$ be a jointly lower-semicontinuous with respect to the product topology on $X \times \mathcal{P}_p(Y)$, bounded from below, and convex in the second argument, cost function for a weak optimal transport problem with marginals $\mu \in \mathcal{P}(X),\ \nu \in \mathcal{P}_p(Y)$. Then,*

$$\inf_{\pi\in Cpl(\mu,\nu)} \int_X C(x,\pi_x)\ d\mu(x) = \sup_{\substack{(\varphi,\psi)\in D(C)\\ \varphi\in C_b(X),\psi\in C_p(Y)}} \int_X \varphi\ d\mu + \int_Y \psi\ d\nu$$

*where $D(C) := \{(\varphi,\psi) : X \times Y \to \mathbb{R} \mid \varphi \oplus \int \psi d\rho \le C,\ |\psi(y)| \le K(1 + d(y,y_0)^p)\}$.
Moreover, duality holds also with C-transforms, i.e.*

$$\inf_{\pi\in Cpl(\mu,\nu)} \int_X C(x,\pi_x)\ d\mu(x) = \sup_{\psi\in C_{b,p}(Y)} \int_X \psi^C\ d\mu + \int_Y \psi\ d\nu,$$

where $\psi^C : X \to \mathbb{R} \cup \{+\infty\}$ such that $\psi^C(x) = \inf_{\rho\in\mathcal{P}_p(Y)} C(x,\rho) - \int \psi\ d\rho$

Proof. Theorem 1.2, [2]. □

Remark 2.2.3. The restriction to the space $\psi \in C_{b,p}(Y)$ is necessary, since the C-transform ψ^C might not be integrable only for $\psi \in C_p(Y)$.

Now, we will consider some weak generalizations of the classical optimal transport theorems in the real finite-dimensional setting $X = Y = \mathbb{R}^d$, namely the Kantorovich-Rubenstein formula and the Brenier theorem.

Lemma 2.2.4. *Let $\varphi : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be lower-semicontinuous, bounded below function. Then,*

$$\varphi^{**}(x) = \inf_{\substack{\rho \in \mathcal{P}_1(\mathbb{R}^d) \\ mean(\rho)=x}} \int \varphi \, d\rho \quad \text{for any } x \in \mathbb{R}^d.$$

Proof. Note that the function $\psi : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ given by

$$\psi(x) = \inf_{\substack{\rho \in \mathcal{P}_1(\mathbb{R}^d) \\ mean(\rho)=x}} \int_{\mathbb{R}^d} \varphi(y) \, d\rho(y)$$

is lower-semicontinuous and convex, since φ lower-semicontinuous and bounded below. Moreover, it holds $\psi(x) \le \varphi(x)$ by taking $\rho = \delta_x$. Hence, ψ is a lower-semicontinuous convex function smaller than φ. Therefore, we have $\psi \le \varphi^{**}$ by Fenchel-Moreau theorem. Viceversa, fix $\rho \in \mathcal{P}_1(\mathbb{R}^d)$ with $mean(\rho) = x$, and obtain

$$\begin{aligned} \varphi^{**}(x) = \varphi^{**}(mean(\rho)) &= \varphi^{**}\left(\int_{\mathbb{R}^d} y \, d\rho(y)\right) \\ &\le \int_{\mathbb{R}^d} \varphi^{**}(y) \, d\rho(y) \quad \text{by Jensen, since } \varphi^{**} \text{ convex} \\ &\le \int_{\mathbb{R}^d} \varphi(y) \, d\rho(y) \quad \text{by } \varphi^{**} \le \varphi. \end{aligned}$$

□

Theorem 2.2.5. *(Kantorovich-Rubenstein) The weak optimal transport problem with cost $C(x, \rho) = |x - mean(\rho)|$ on $\mathbb{R}^d$, and marginals $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the duality*

$$\inf_{\pi \in Cpl(\mu,\nu)} \int C(x, \pi_x) \, d\mu(x) = \sup_{\varphi \text{ 1-Lip, concave}} \int \varphi \, d(\nu - \mu).$$

Proof. Note that the weak cost function $C(x, \rho) = |x - mean(\rho)|$ is continuous, non-negative and convex in the second argument, then we get by duality theorem

$$\inf_{\pi \in Cpl(\mu,\nu)} \int C(x, \pi_x) \, d\mu(x) = \sup_{\psi \in C_{b,1}(\mathbb{R}^d)} \int_{\mathbb{R}^d} \psi^C \, d\mu + \int_{\mathbb{R}^d} \psi \, d\nu.$$

Moreover, note that the C-transform of measurable map $\psi : \mathbb{R}^d \to \mathbb{R}$ such that $\psi \in$

$C_{b,1}(\mathbb{R}^d)$ is given by

$$
\begin{aligned}
\psi^C(x) &= \inf_{\rho \in \mathcal{P}_1(\mathbb{R}^d)} |x - mean(\rho)| - \int \psi \ d\rho \\
&= \inf_{z \in \mathbb{R}^d} \inf_{\substack{\rho \in \mathcal{P}_1(\mathbb{R}^d) \\ mean(\rho) = z}} |x - z| - \int \psi \ d\rho \\
&= \inf_{z \in \mathbb{R}^d} \left(|x - z| + \inf_{\substack{\rho \in \mathcal{P}_1(\mathbb{R}^d) \\ mean(\rho) = z}} \int -\psi \ d\rho \right) \\
&= \inf_{z \in \mathbb{R}^d} |x - z| + (-\psi)^{**}(z),
\end{aligned}
$$

by Lemma 2.2.4. In particular, if ψ is also concave, then $-\psi$ is convex and $(-\psi)^{**} = -\psi$ by Fenchel-Moreau theorem. Hence, we obtain

$$
\psi^C(x) = \inf_{z \in \mathbb{R}^d} |x - z| - \psi(z) = \psi^c(x) = -\psi(x),
$$

since $\psi \in C_{b,1}(\mathbb{R}^d)$ is 1-Lipschitz, where $c(x, y) = |x - y|$ metric cost in $\mathbb{R}^d$. Therefore, we obtain

$$
\begin{aligned}
\inf_{\pi \in Cpl(\mu,\nu)} \int C(x, \pi_x) \ d\mu(x) &= \sup_{\psi \in C_{b,1}(\mathbb{R}^d)} \int_{\mathbb{R}^d} \psi^C \ d\mu + \int_{\mathbb{R}^d} \psi \ d\nu \\
&= \sup_{\psi \ \text{1-Lip, concave}} \int -\psi \ d\mu + \int \psi \ d\nu.
\end{aligned}
$$

□

Now, we give a weak version of the Brenier theorem on $X = Y = \mathbb{R}^d$ with quadratic cost, firstly proved in 2019 by [2].

Definition 2.2.6. A probability $\mu \in \mathcal{P}_1(\mathbb{R}^d)$ is in convex order with a probability $\nu \in \mathcal{P}_1(\mathbb{R}^d)$, and we write $\mu \leq_c \mu$, if

$$
\int f \ d\mu \leq \int f \ d\nu
$$

for any $f : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ convex.

Theorem 2.2.7. *(Brenier-Strassen) Let $X = Y = \mathbb{R}^d$ with cost $C(x, \rho) = |x - mean(\rho)|^2$, and marginals $\mu \in \mathcal{P}_2(\mathbb{R}^d), \nu \in \mathcal{P}_1(\mathbb{R}^d)$. Then, there exists a unique $\mu^* \in \mathcal{P}_2(\mathbb{R}^d)$ of the form $\mu^* = \nabla\varphi(\mu)$ for some $\varphi : \mathbb{R}^d \to \mathbb{R}$ 1-Lipschitz, such that*

$$
\mu \leq_C \nu,
$$

and

$$
\mathcal{W}_2(\mu^*, \mu)^2 = \inf_{\eta \leq_C \nu} \mathcal{W}_2(\eta, \nu)^2 = \inf_{\pi \in Cpl(\mu,\nu)} \int |x - mean(\pi_x)|^2 \ d\mu(x).
$$

Moreover, the weak optimal problem V_C^{WOT} admits optimizers, and a coupling $\pi \in Cpl(\mu, \nu)$ is optimal if and only if $mean(\pi_x) = \nabla\varphi(x)$ μ-a.s.

2.3 Martingale Optimal Transport (MOT)

In this section, we will always consider the case $X = Y = \mathbb{R}^d$.

Definition 2.3.1. A martingale coupling of two probability measures $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$ is a coupling $\pi \in Cpl(\mu, \nu)$ such that

$$mean(\pi_x) = x \quad \text{for } \mu\text{-a.s. } x \in \mathbb{R}^d,$$

where $mean(\rho) : \mathcal{P}_1(\mathbb{R}^d) \to \mathbb{R}^d$ such that $\text{mean}(\rho) = \int_{\mathbb{R}^d} x \, d\rho(x)$.

We will denote the space of martingale couplings between μ and ν by $Cpl^M(\mu, \nu)$.

Example 2.3.2.

(i) *Martingale Monge Couplings*
Let $\mu \in \mathcal{P}_1(\mathbb{R}^d)$, then the Monge coupling $\pi := (Id, Id)\mu \in Cpl(\mu, \mu)$ is a martingale coupling. Indeed, we have

$$mean(\pi_x) = x \quad \mu\text{-a.s.}$$

Viceversa, if $\pi := (Id, T)\mu \in Cpl(\mu, \nu)$ is a Monge coupling between μ, ν induced by a transport map T, and π is a martingale coupling, then

$$x = mean(\pi_x) = T(x) \quad \mu\text{-a.s.}$$

(ii) *Martingale Dirac Couplings*
Let $\delta_x \in \mathcal{P}_1(\mathbb{R}^d)$ be a Dirac measure, and $\nu \in \mathcal{P}_1(\mathbb{R}^d)$ with $mean(\nu) = x$. Then, the unique martingale coupling between δ_x and ν is $\delta_x \otimes \nu$. In particular, it holds

$$Cpl(\delta_x, \nu) = Cpl^M(\delta_x, \nu) = \{\delta_x \otimes \nu\}.$$

Definition 2.3.3. The martingale optimal transport functional with cost $c : X \times Y \to \mathbb{R} \cup \{+\infty\}$ is the map $V_c^{MOT} : \mathcal{P}_1(\mathbb{R}^d) \times \mathcal{P}_1(\mathbb{R}^d) \to \mathbb{R} \cup \{+\infty\}$ such that

$$V_c^{MOT}(\mu, \nu) := \inf_{\pi \in Cpl^M(\mu,\nu)} \int_{X \times Y} c \, d\pi.$$

Remark 2.3.4. The martingale optimal transport problem can be interpreted as a weak optimal transport problem with degenerate cost function

$$C(x, \rho) := \begin{cases} \int_Y c(x, y) \, d\rho(y) & mean(\rho) = x \\ +\infty & \text{otherwise.} \end{cases}$$

A natural first question is whether the martingale transport problem is well posed. In the classical Kantorovich formulation of optimal transport, a fundamental property is that the set of couplings $Cpl(\mu, \nu)$ is always non-empty. One may therefore ask whether an analogous existence result holds in the martingale setting. Strassen's theorem characterizes the existence of martingale couplings.

Theorem 2.3.5. *(Strassen) Let $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$, then $Cpl^M(\mu, \nu) \neq \varnothing \iff \mu \leq_c \nu$.*

Proof. Fix $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$, and note that there exists a martingale coupling between μ and ν if and only if

$$V_C^{WOT}(\mu, \nu) = \inf_{\pi \in Cpl(\mu,\nu)} \int_{\mathbb{R}^d} C(x, \pi_x)\ d\mu(x) = 0$$

for the degenerate cost function

$$C(x, \rho) = \begin{cases} 0 & mean(\rho) = x \\ +\infty & \text{otherwise.} \end{cases}$$

Moreover, the weak cost C is a jointly lower-semicontinuous, bounded below and convex in $\mathcal{P}_1(\mathbb{R}^d)$, and we get by Theorem 2.2.2

$$V_C^{WOT}(\mu, \nu) = \sup_{\psi \in C_{b,1}(\mathbb{R}^d)} \int_{\mathbb{R}^d} \psi^C\ d\mu + \int_{\mathbb{R}^d} \psi\ d\nu.$$

Furthermore, the C-transform of ψ is given by

$$\begin{aligned} \psi^C(x) &= \inf_{\rho \in \mathcal{P}_1(\mathbb{R}^d)} C(x, \rho) - \int_{\mathbb{R}^d} \psi(y)\ d\rho(y) \\ &= \inf_{\substack{\rho \in \mathcal{P}_1(\mathbb{R}^d) \\ mean(\rho)=x}} 0 - \int_{\mathbb{R}^d} \psi(y)\ d\rho(y) \\ &= (-\psi)^{**}(x) \quad \text{for all } x \in \mathbb{R}^d, \end{aligned}$$

by Lemma 2.2.4, since $\psi \in C_{b,1}(\mathbb{R}^d)$ is lower-semicontinuous and bounded below.
Finally, we get

$$\begin{aligned} V_C^{WOT}(\mu, \nu) &= \sup_{\psi \in C_{b,1}(\mathbb{R}^d)} \int_{\mathbb{R}^d} \psi^C\ d\mu + \int_{\mathbb{R}^d} \psi\ d\nu \\ &= \sup_{\psi \in C_{b,1}(\mathbb{R}^d)} \int_{\mathbb{R}^d} (-\psi)^{**}\ d\mu - \int_{\mathbb{R}^d} (-\psi)\ d\nu \\ &\geq \sup_{\psi \in C_{b,1}(\mathbb{R}^d)} \int_{\mathbb{R}^d} (-\psi)^{**}\ d\mu - \int_{\mathbb{R}^d} (-\psi)^{**}\ d\nu \quad \text{since } (-\psi)^{**} \leq -\psi \\ &\geq \sup_{f \text{ convex}} \int_{\mathbb{R}^d} f\ d\mu - \int_{\mathbb{R}^d} f\ d\nu, \end{aligned}$$

since the biconjugate $(-\psi)^{**}$ is always a convex function. Then, there exists a martingale coupling between μ and ν if and only if

$$\sup_{f \text{ convex}} \int_{\mathbb{R}^d} f\ d\mu - \int_{\mathbb{R}^d} f\ d\nu \leq 0,$$

which is equivalent to the convex ordering of μ, ν. □

Once non-triviality is established, we are interested in the topological properties of the space of martingale couplings $Cpl^M(\mu, \nu)$.
Note that, in the following, we will consider only the case $d = 1$.

Theorem 2.3.6. *Let $\mu, \nu \in \mathcal{P}_1(\mathbb{R})$. Then, $Cpl^M(\mu, \nu) \subseteq Cpl(\mu, \nu)$ convex, compact.*

Proof. □

Once again, we are interested in extending the classical results of optimal transport to the martingale optimal transport.

Theorem 2.3.7. *(Existence) Let $c : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a LSC, bounded below cost function for a martingale optimal transport problem with marginals $\mu, \nu \in \mathcal{P}_1(\mathbb{R})$. Then, there exists an optimal martingale transport plan, i.e.*

$$\exists\ \pi^* \in argmin_{\pi \in Cpl^M(\mu,\nu)} \int c\ d\pi.$$

Theorem 2.3.8. *(Duality) Let $c : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a LSC, bounded below cost function for a martingale optimal transport problem with marginals $\mu, \nu \in \mathcal{P}_1(\mathbb{R})$. Then, duality holds, i.e.*

$$\inf_{\pi \in Cpl^M(\mu,\nu)} \int c\ d\pi = \sup_{\substack{f,g,h \in C_b(\mathbb{R}) \\ f(x)+g(y)+h(x)(y-x) \leq c(x,y)}} \left(\int_{\mathbb{R}} f\ d\mu + \int_{\mathbb{R}} g\ d\nu \right).$$

Remark 2.3.9. Martingale duality may not hold.
For example, consider $X = Y = [0, 1]$ with (non lower-semicontinuous) cost

$$c(x, y) = \begin{cases} 1 & x = y \\ 0 & \text{otherwise.} \end{cases}$$

Then, duality does not hold for the martingale problem with marginals $\mu = \nu = \lambda_{|[0,1]}$.

Definition 2.3.10. A couple $\mu, \nu \in \mathcal{P}_1(\mathbb{R})$ is irreducible if

1. $\mu \leq_C \nu$
2. There exists a martingale coupling $\pi^* \in Cpl^M(\mu, \nu)$ such that, for all $A, B \in \mathcal{B}(\mathbb{R})$

$$\mu(A),\ \nu(B) > 0 \implies \pi(A \times B) > 0.$$

Theorem 2.3.11. *(Optimality) Let $\mu, \nu \in \mathcal{P}_1(\mathbb{R})$ irreducible, compactly supported, and let $c : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a continuous, bounded cost function. Then,*

$$\inf_{\pi \in Cpl^M(\mu,\nu)} \int c\ d\pi = \int f\ d\mu + \int f\ d\nu$$

for some $f \in L^1(\mu), g \in L^1(\nu), h \in C(\mathbb{R})$.

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